DebCovDiff: Differential Testing of Coverage Measurement Tools on Real-World Projects

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Code Coverage 101

Function under test:

Test cases:

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int f(int a, int b, int c) {
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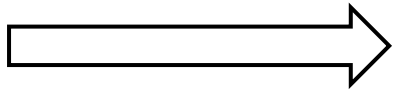

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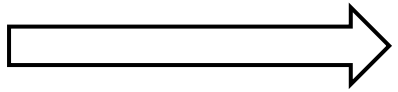

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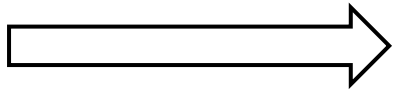

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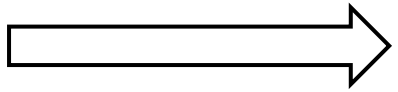

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How many times is each line executed?

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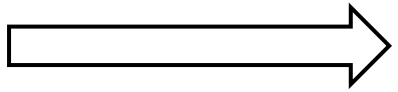

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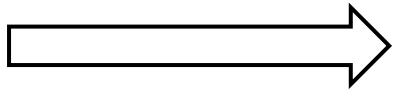

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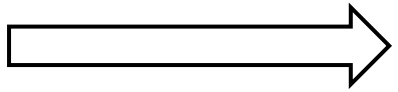

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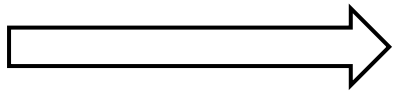

```
3 | int f(int a, int b, int c) {
3 |     if ( (a && b) || c )
    [True: 2, False: 1]
    [True: 0, False: 2]
    [True: 1, False: 2]
1 |         return 3;
2 |     return 4;
3 | }
```

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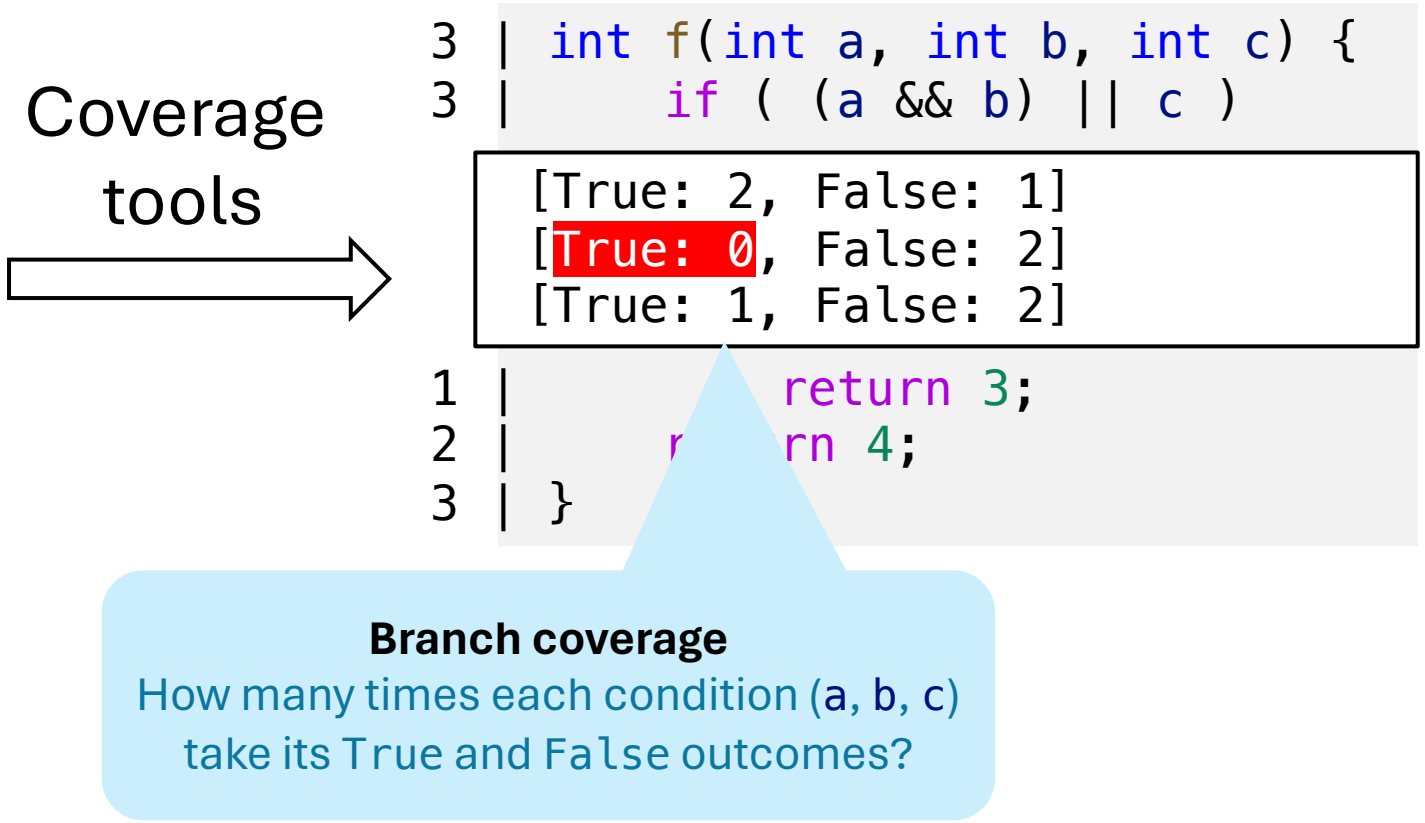

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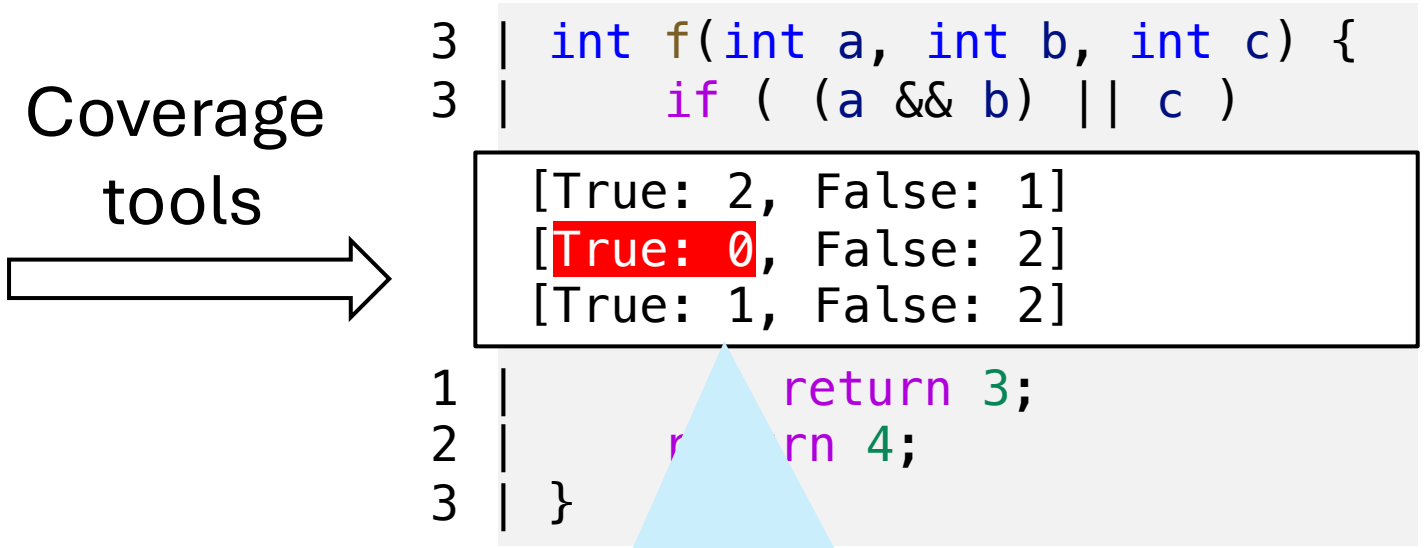

Branch coverage

How many times each condition (a, b, c) take its True and False outcomes?

Some missed

Code Coverage 101

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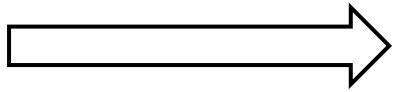

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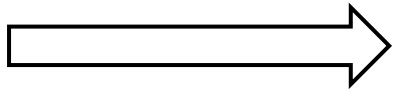

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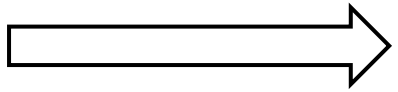

```
3 | int f(int a, int b, int c) {
3 |     if ( (a && b) || c )

     C1, C2, C3   Result
 1 { F , - , F  = F     }
 2 { T , F , F  = F     }
 3 { T , F , T  = T     }

 C1-Pair: not covered
 C2-Pair: not covered
 C3-Pair: covered: (2,3)

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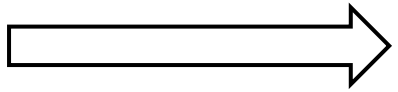

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Does each condition take both outcomes, while the rest of conditions stay the same, and the result differs?

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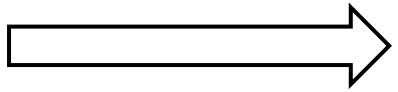

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More missed

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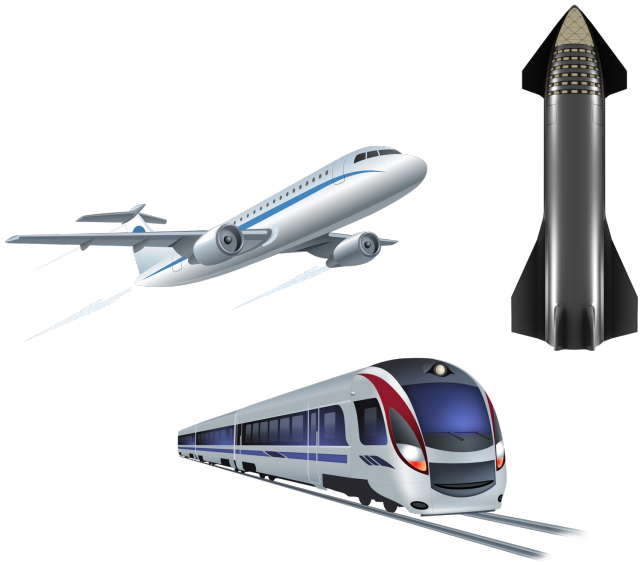

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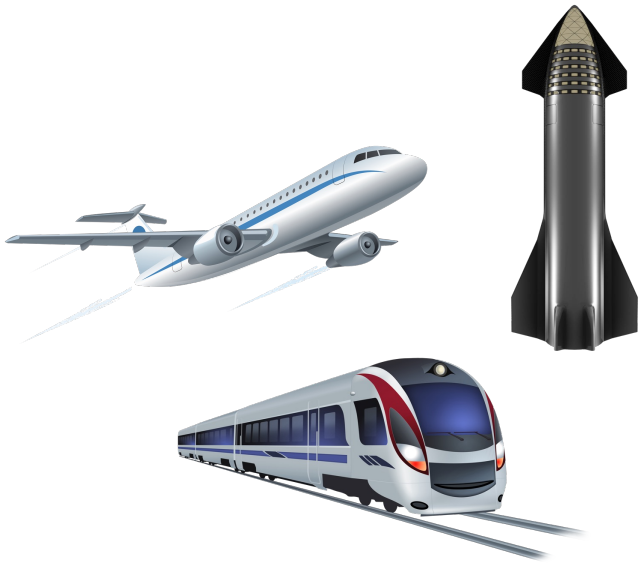

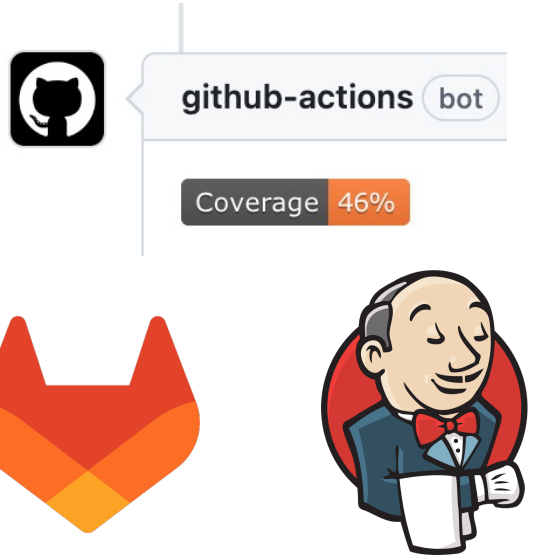

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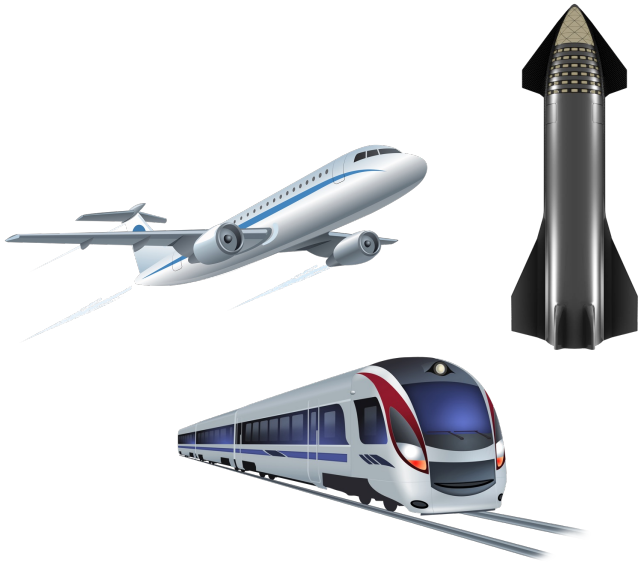

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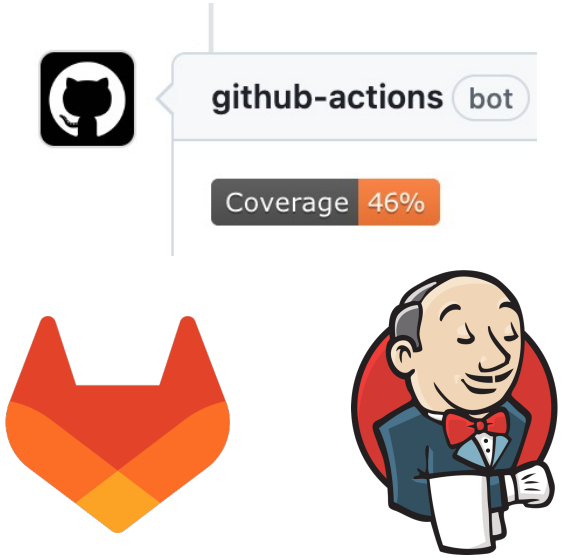

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- Advanced **program analysis**
  - Fuzzing (AFL, `libFuzzer`)
  - Fault localization [Jones et al. 2005]

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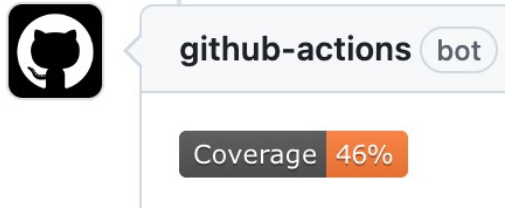

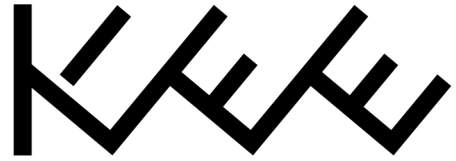

...only when the reported coverage is *correct*!

critical industries

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In Reality, Coverage Tools Can Be Wild…

An LLVM-cov Bug Found in Our Work

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Snippet under test:

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```
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if (family == AF_INET6) {
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    if (IN6_IS_ADDR_LINKLOCAL(&sin6->sin6_addr) ||
        IN6_IS_ADDR_MC_LINKLOCAL(&sin6->sin6_addr))
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https://sources.debian.org/src/hostname/3.23%2Bnmu1/hostname.c#L311

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My-Ubuntu-Laptop
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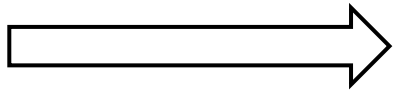

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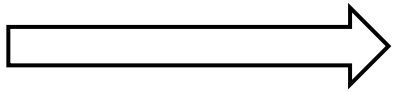

Coverage report for `hostname.c` by LLVM-cov

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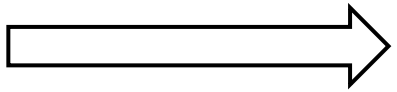

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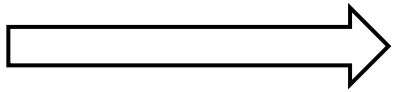

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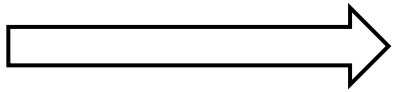

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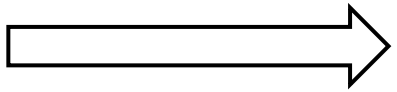

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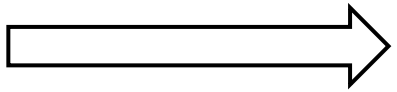

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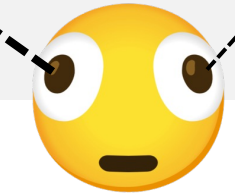

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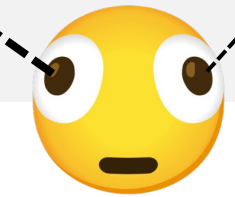

$1.84\text{E} = 1.84\times10^{18}$

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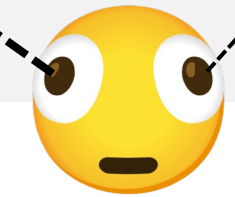

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An Intel Core i9-14900k (5 GHz) would need **117 years** to print your hostname!

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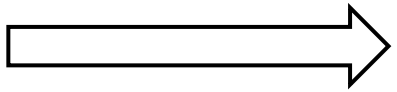

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You can play with it on your phone!
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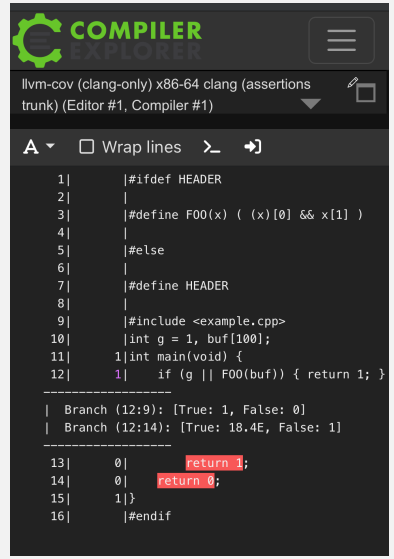

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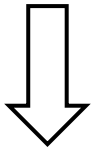

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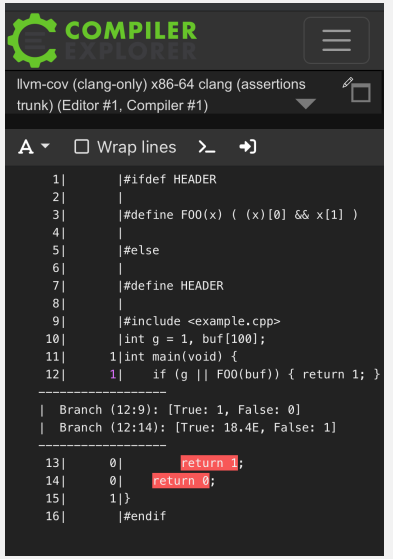

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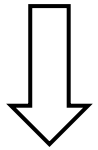

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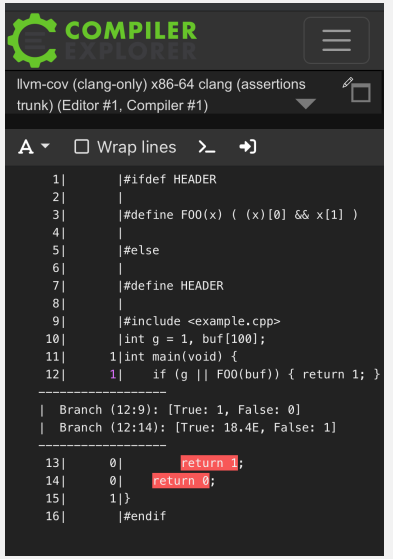

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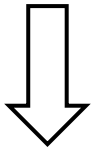

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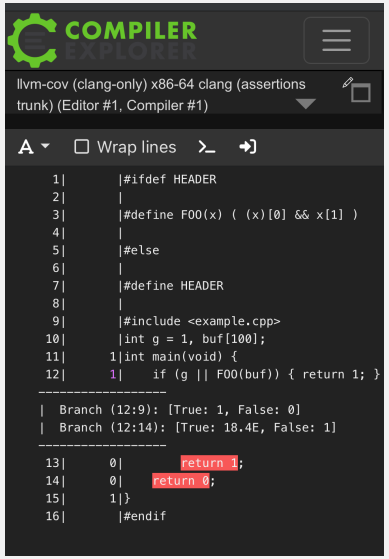

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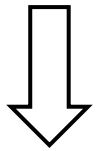

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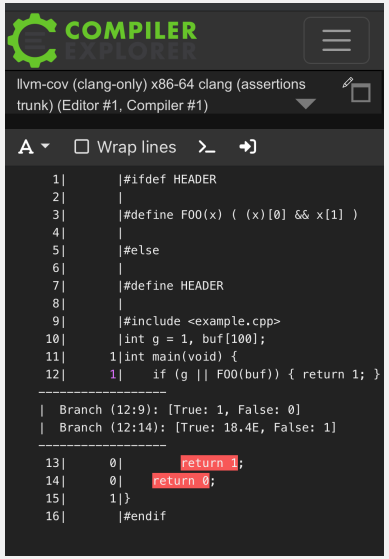

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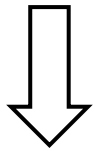

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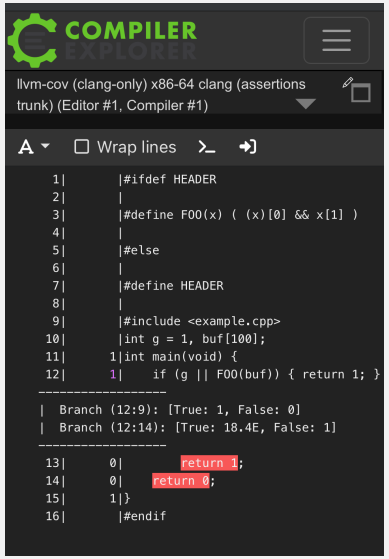

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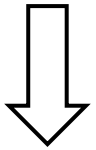

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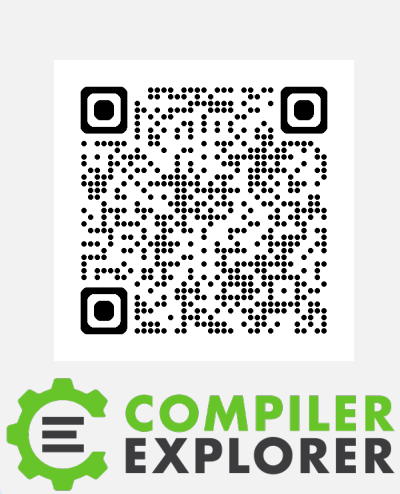

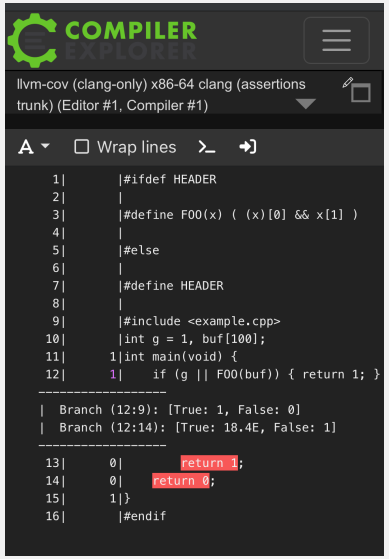

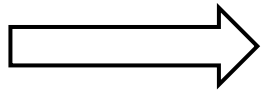

③ Reported as LLVM#131505 by our work

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Though complex, not capturing **real-world** code patterns

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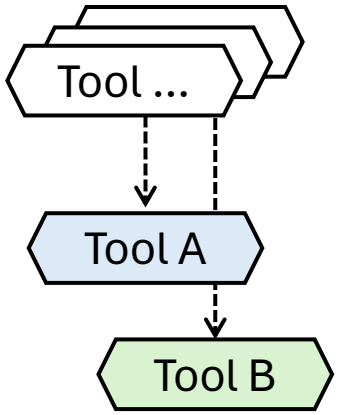

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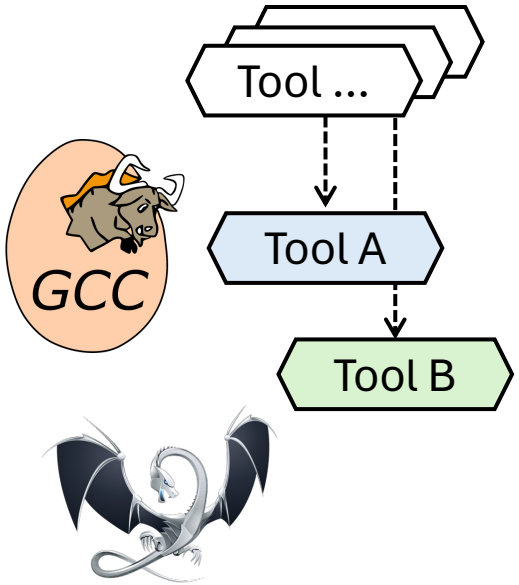

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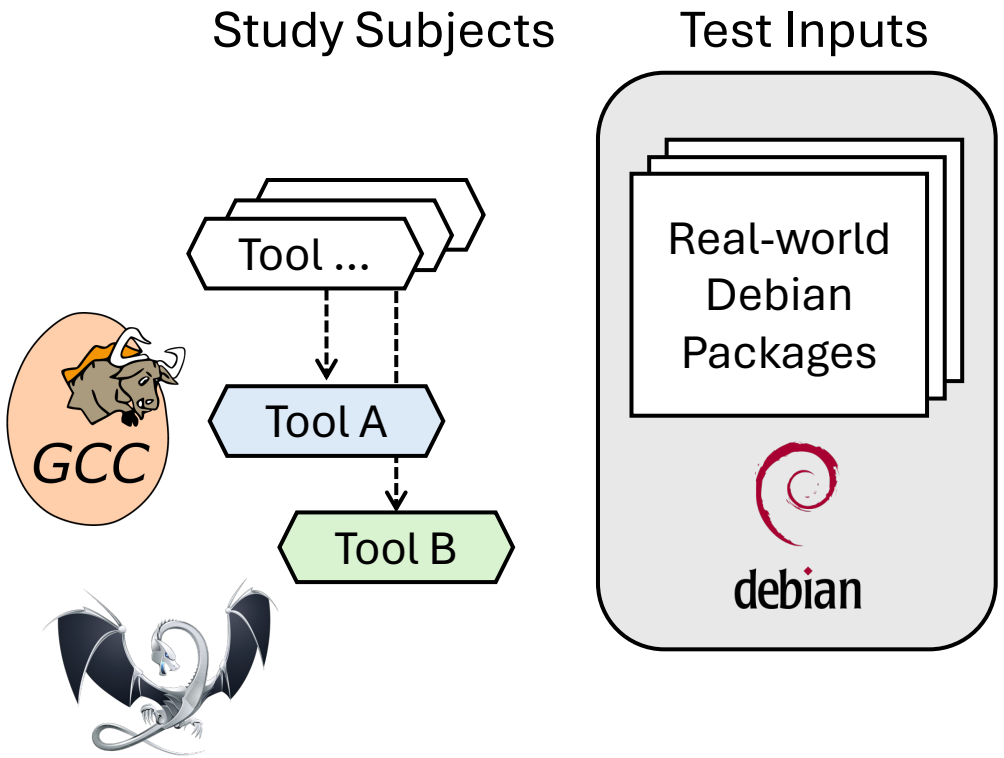

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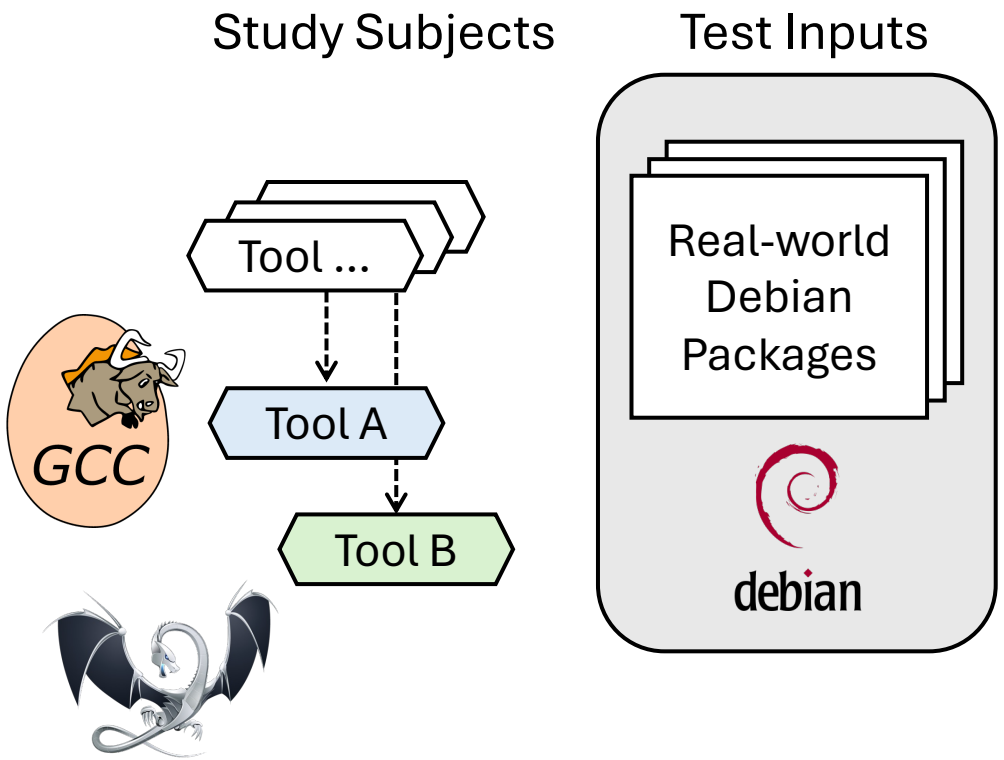

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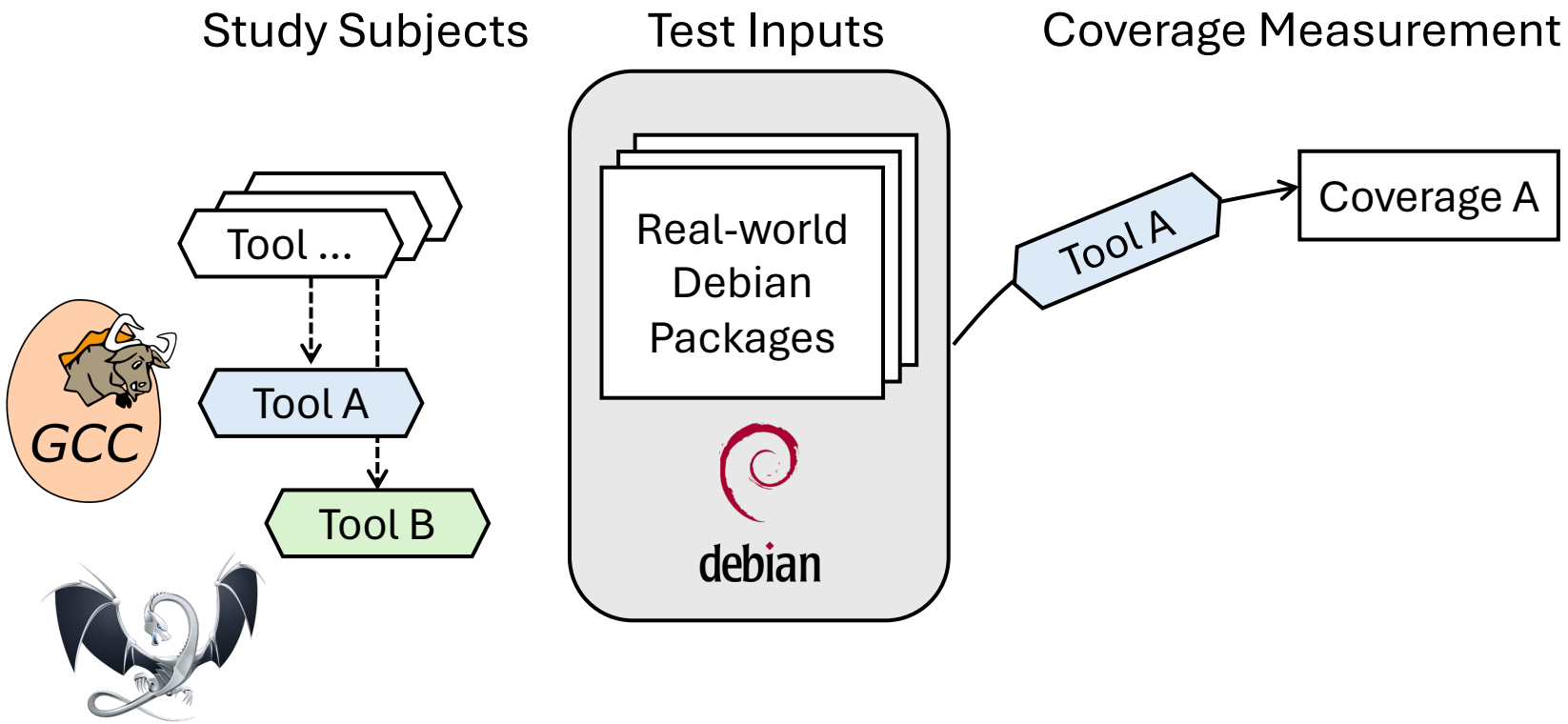

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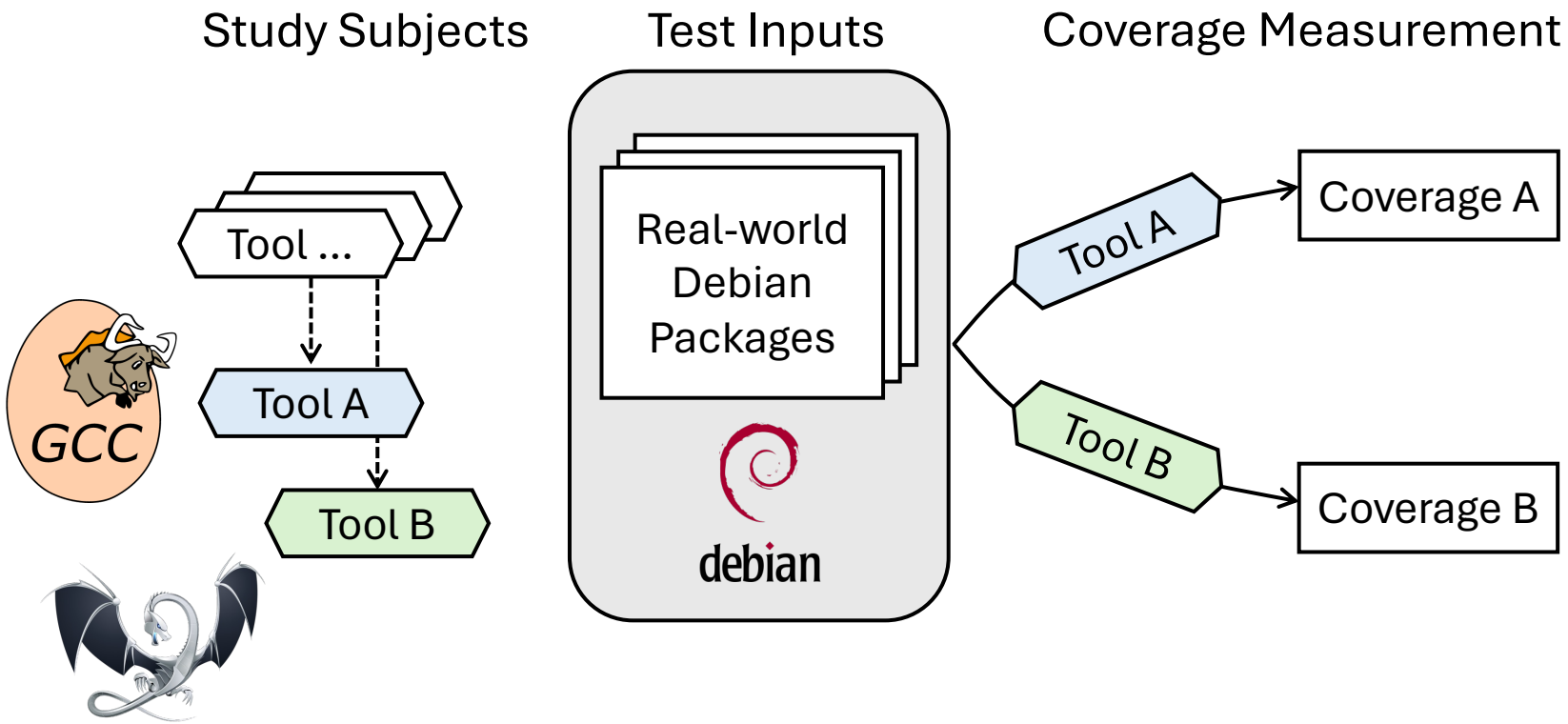

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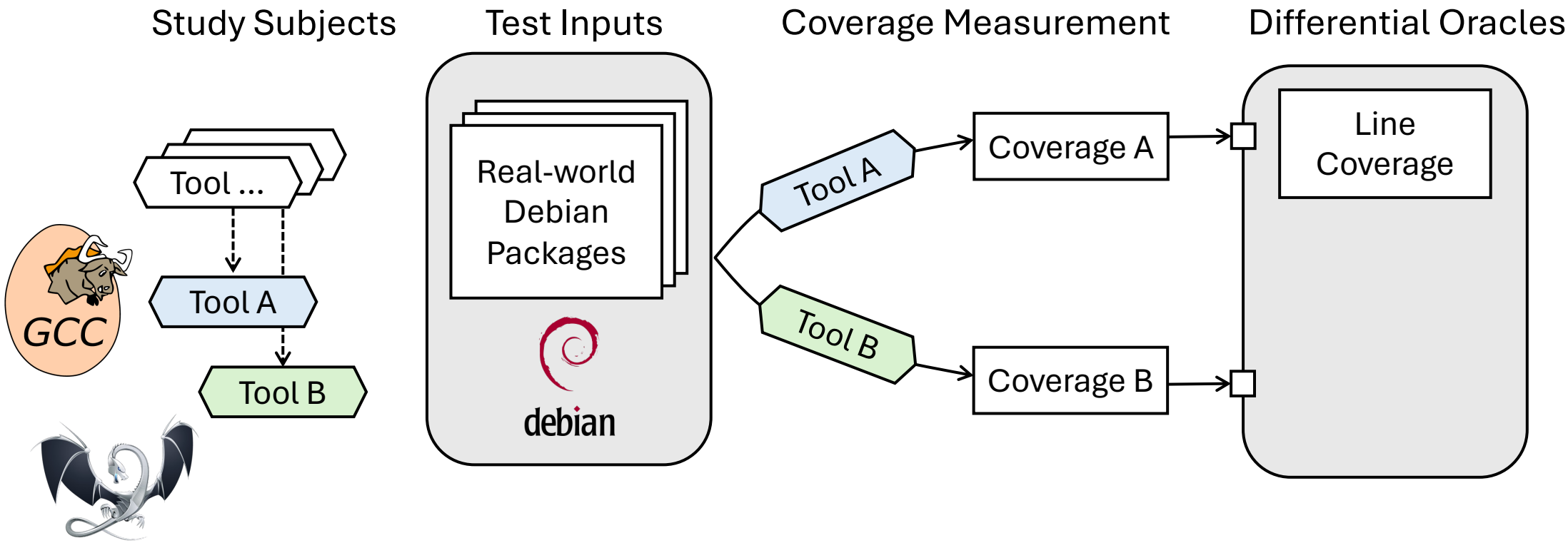

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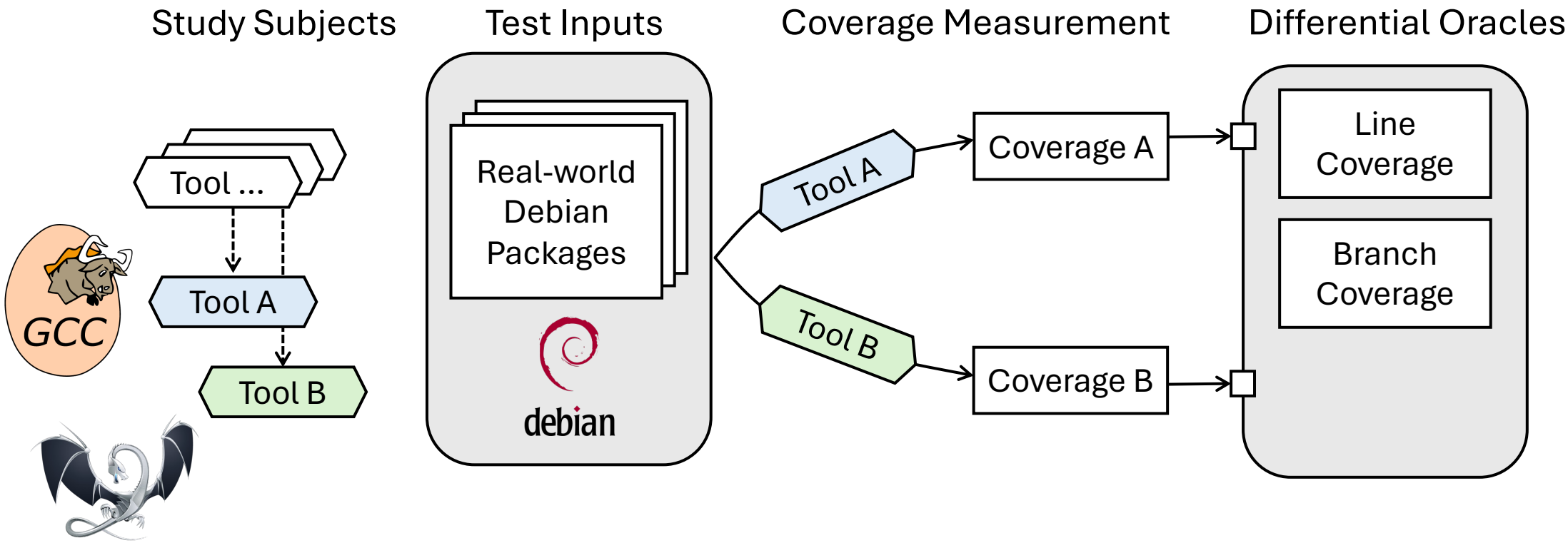

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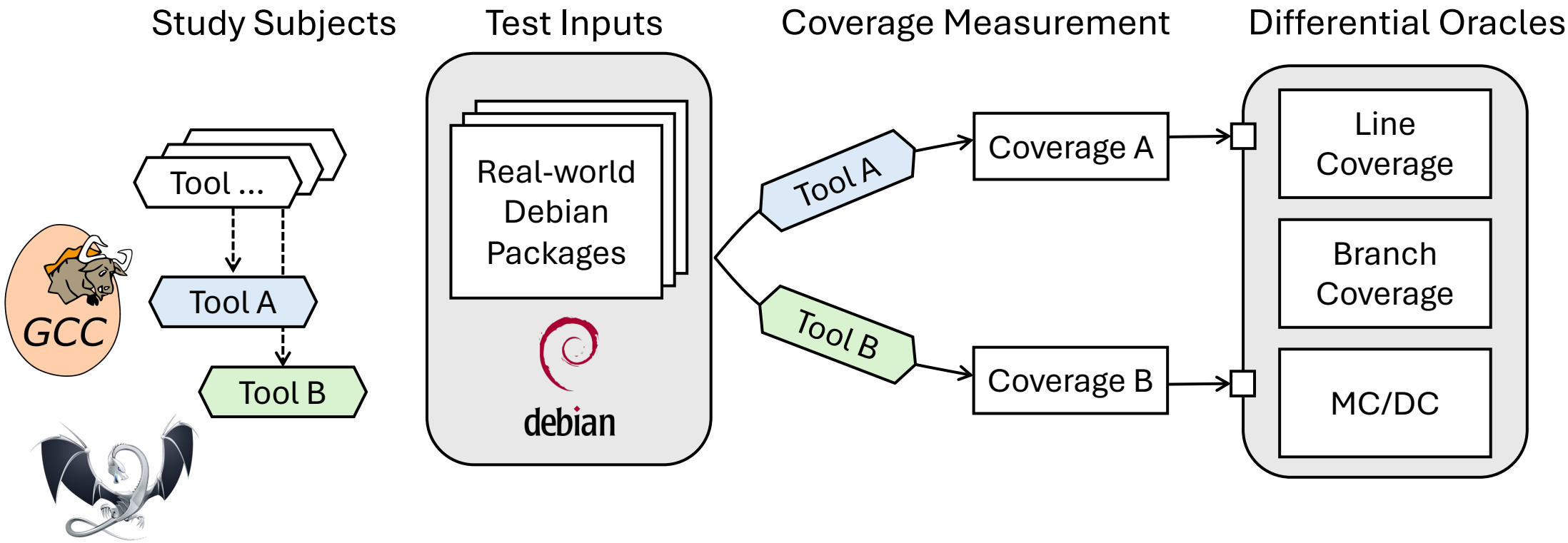

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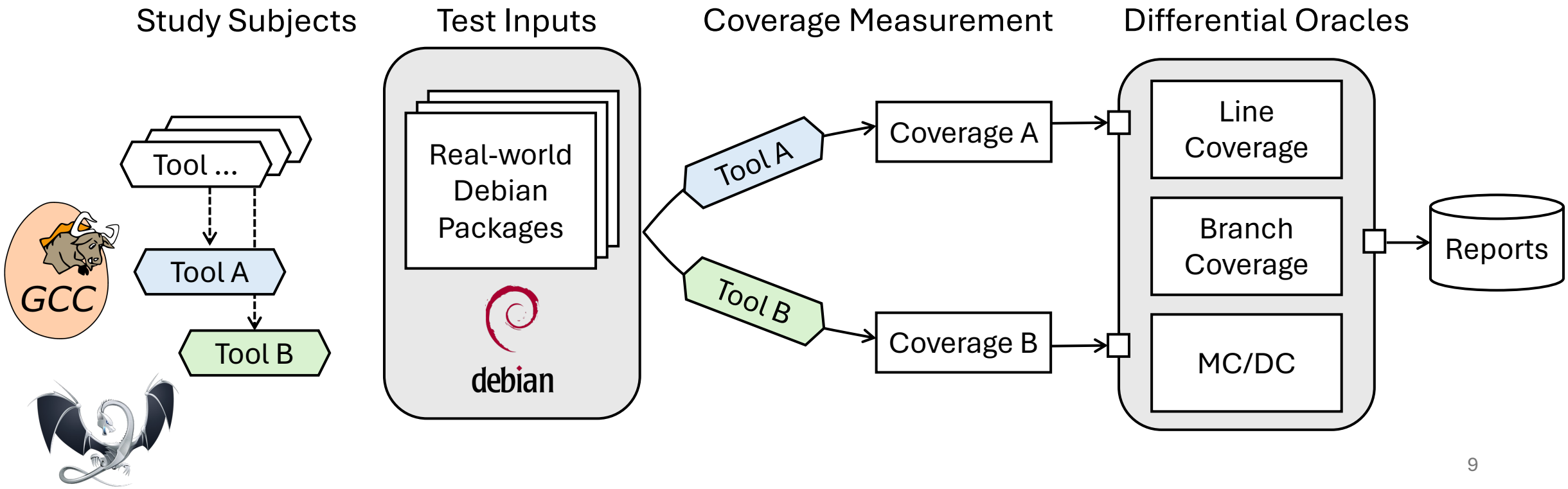

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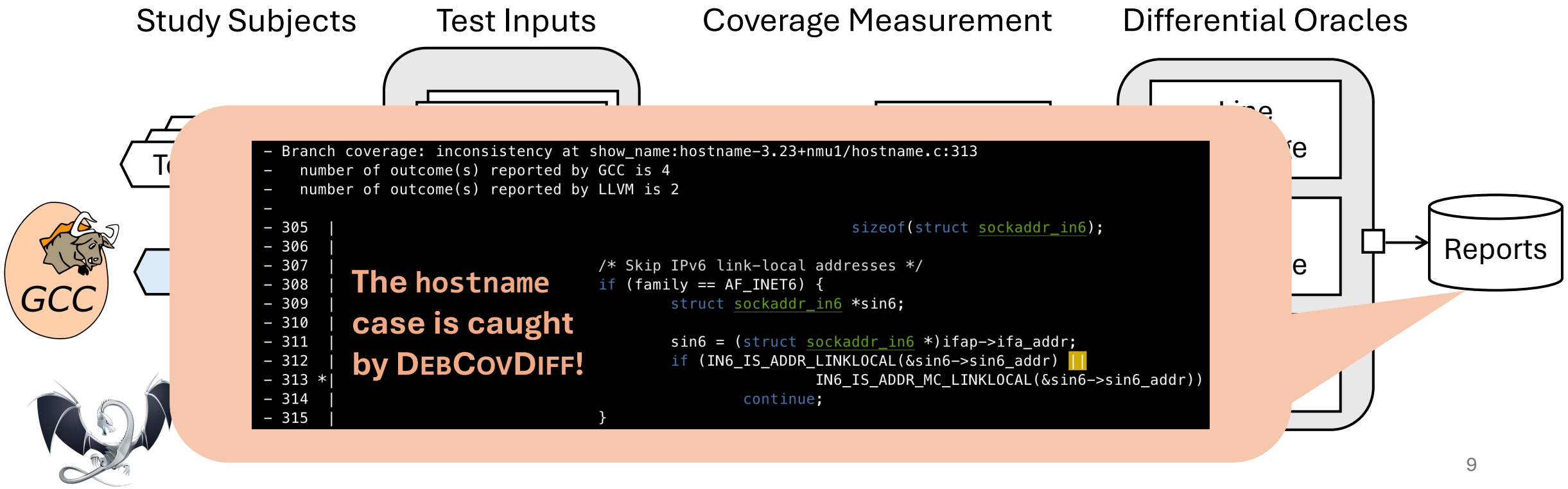

Debian Packages as Test Input

Selection criteria:
relevance and heterogeneity

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All Debian 12.8 (“bookworm”) binary packages for **amd64** under **“main”** component

63,417

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Package examples

- Essential system utilities: **`coreutils`**
- Networking tools: **`iproute2`**
- Compression libraries: **`bzip2`**
- Services & management: **`apache2`**
- User-facing tools: **`nano`**

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Goal: execute *some* code paths in Debian packages so that coverage comparison is more meaningful

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Differential Oracles

- Line coverage

```
3 |
3 | (C/C++ source line...)
3 |
```

Gcov report

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LLVM-cov report

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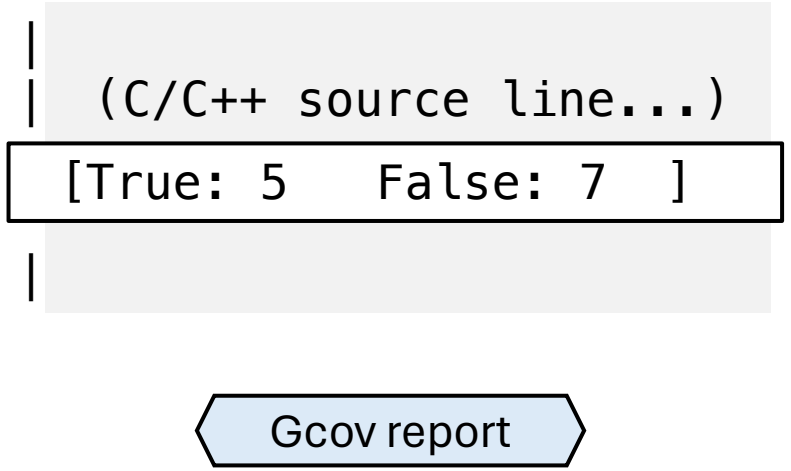

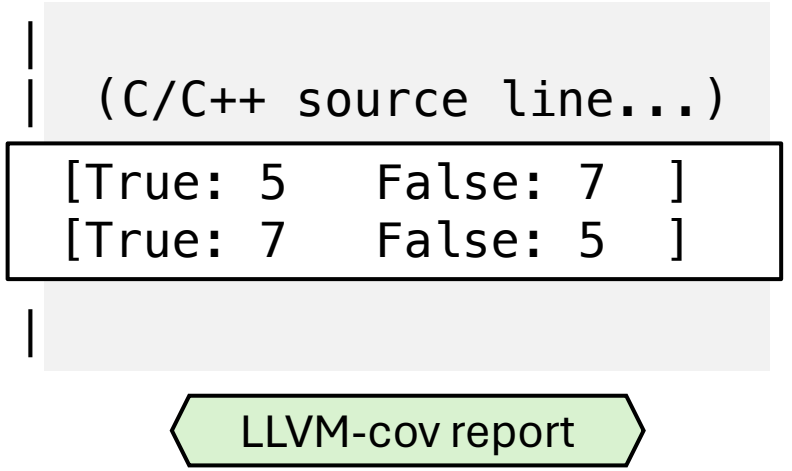

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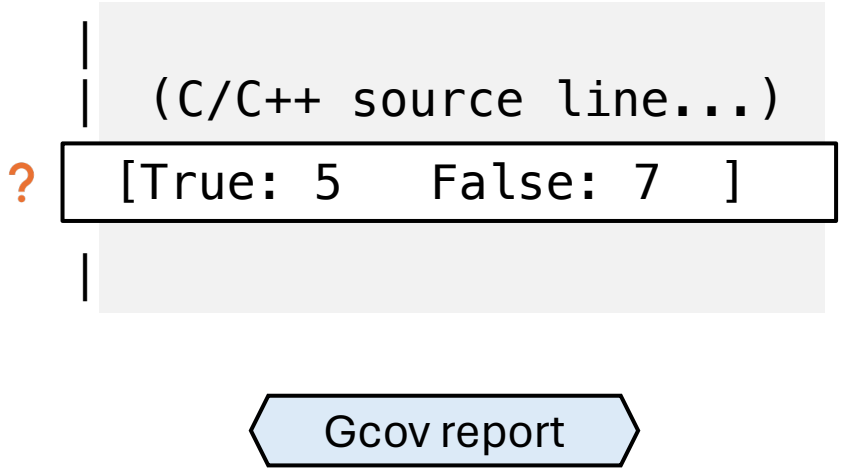

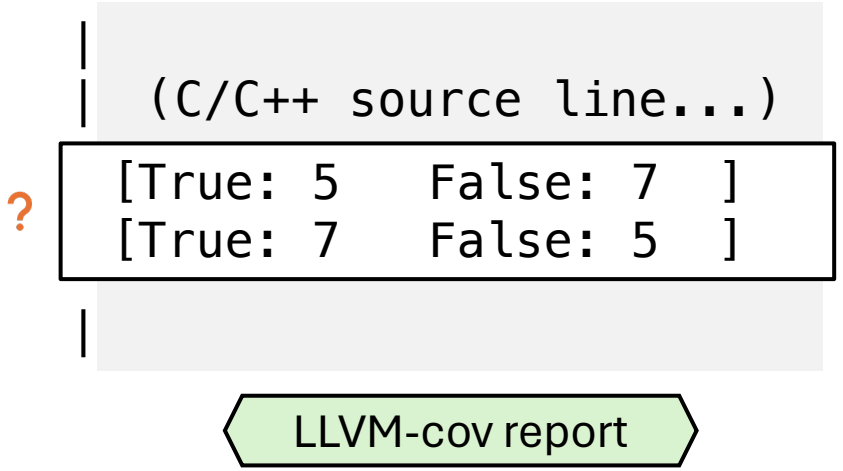

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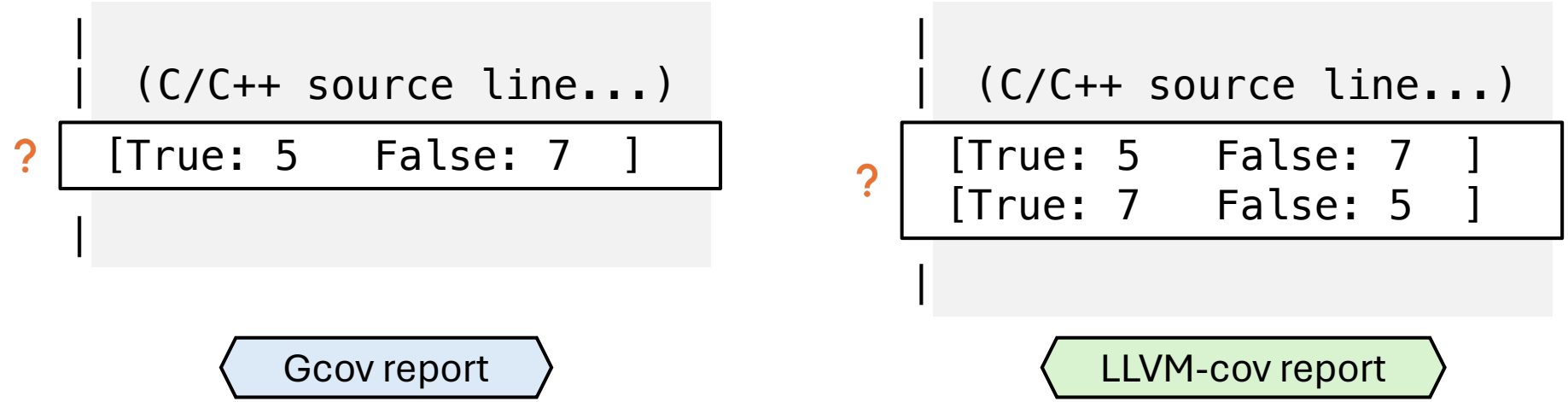

Numbers of branch conditions are already different!

Differential Oracles

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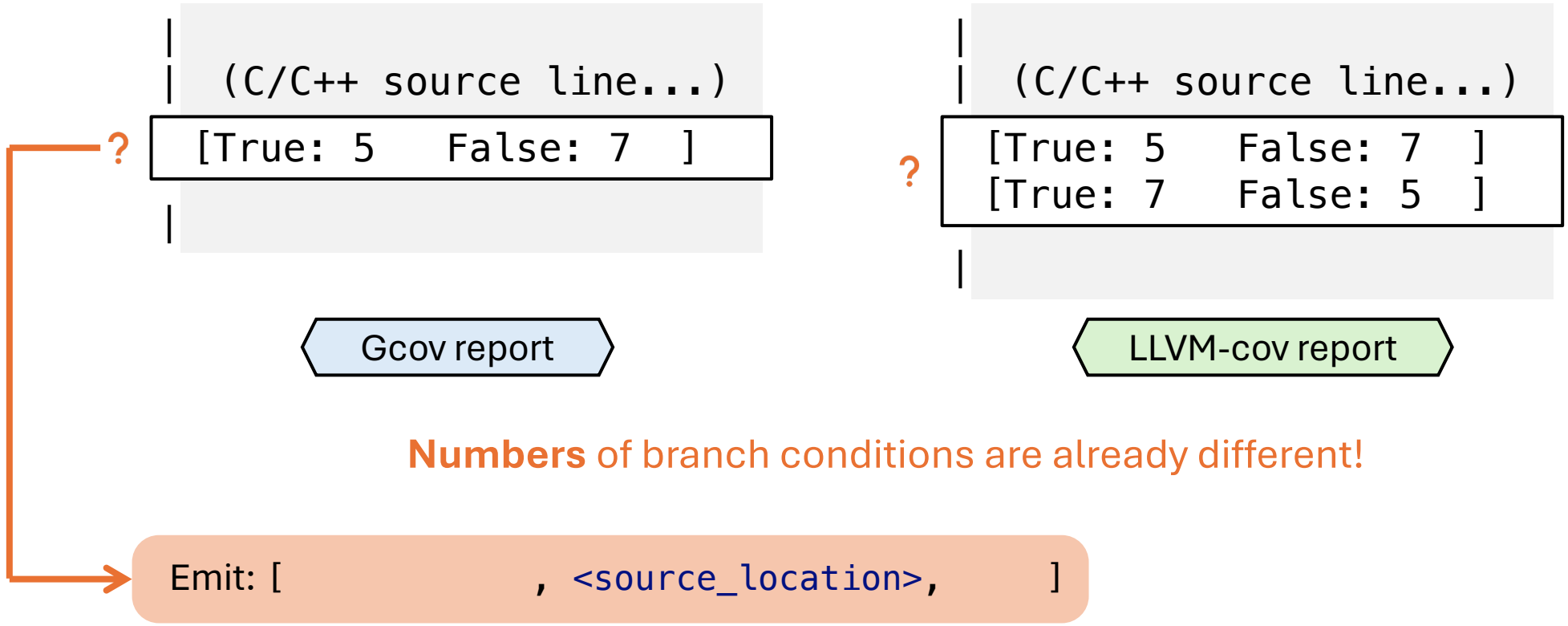

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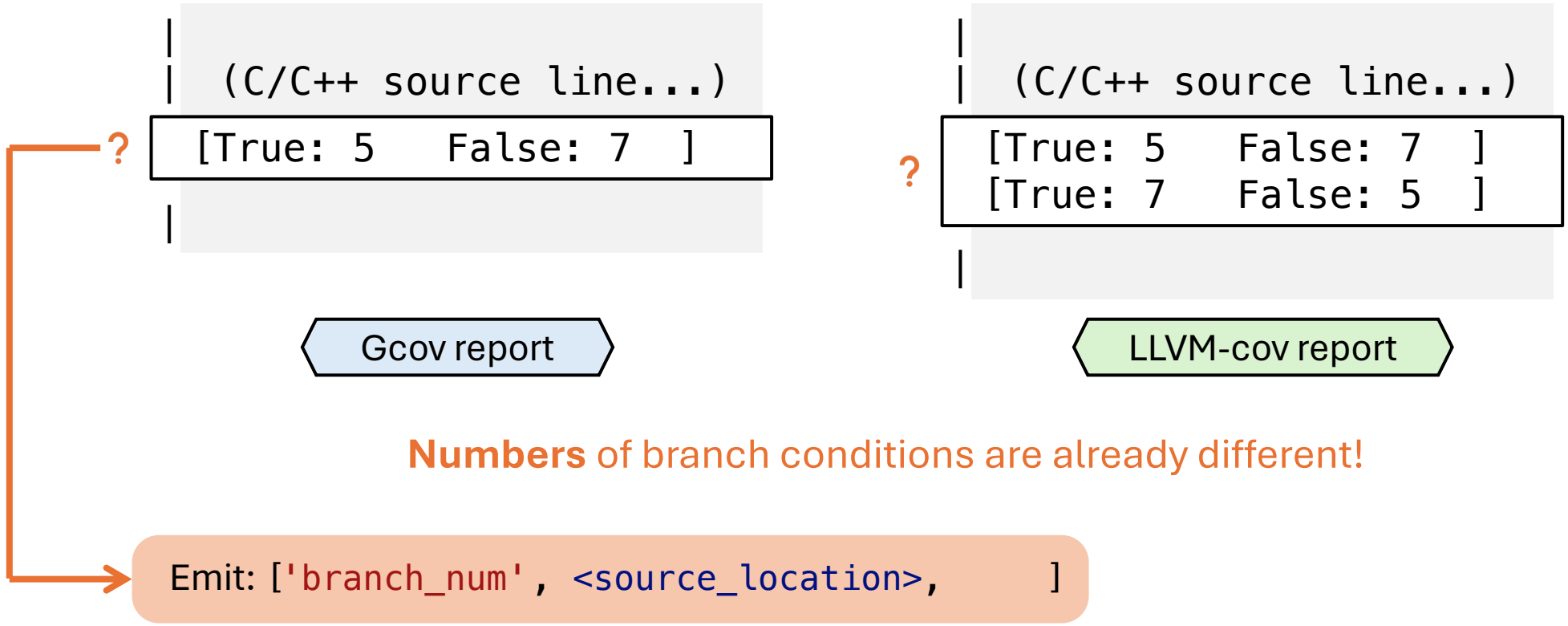

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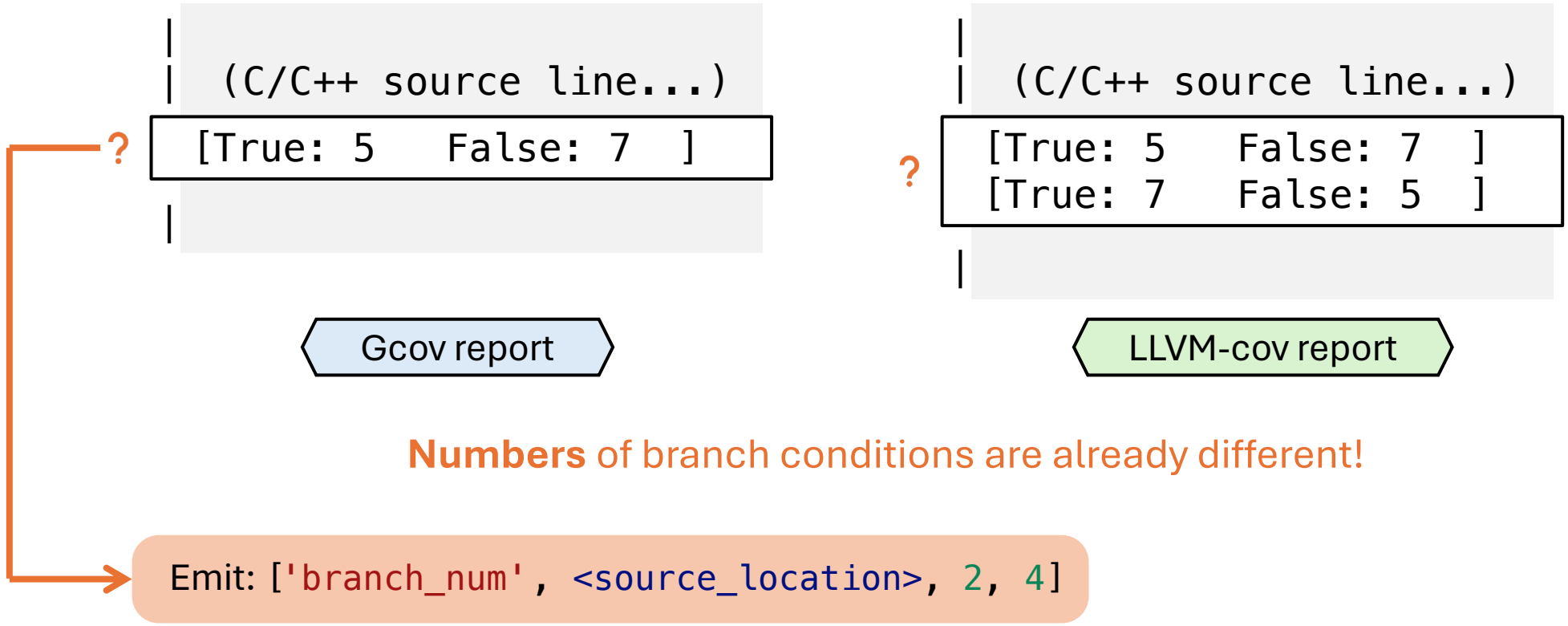

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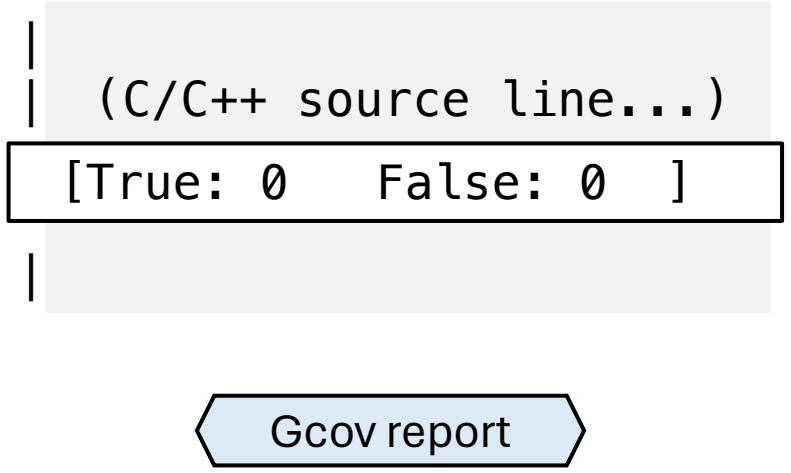

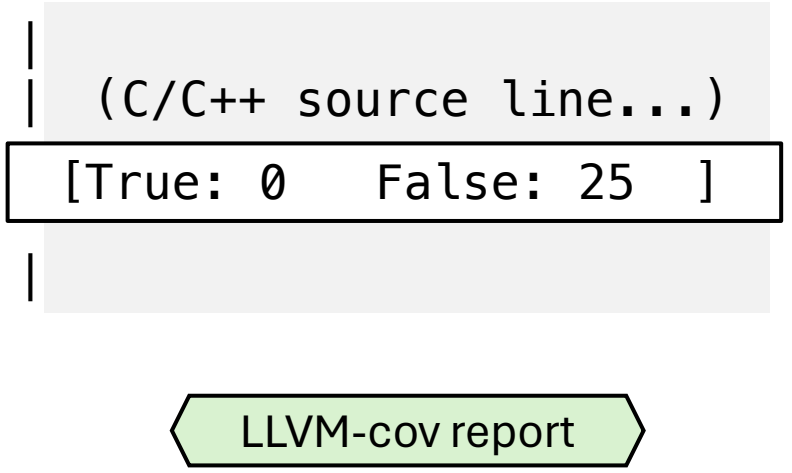

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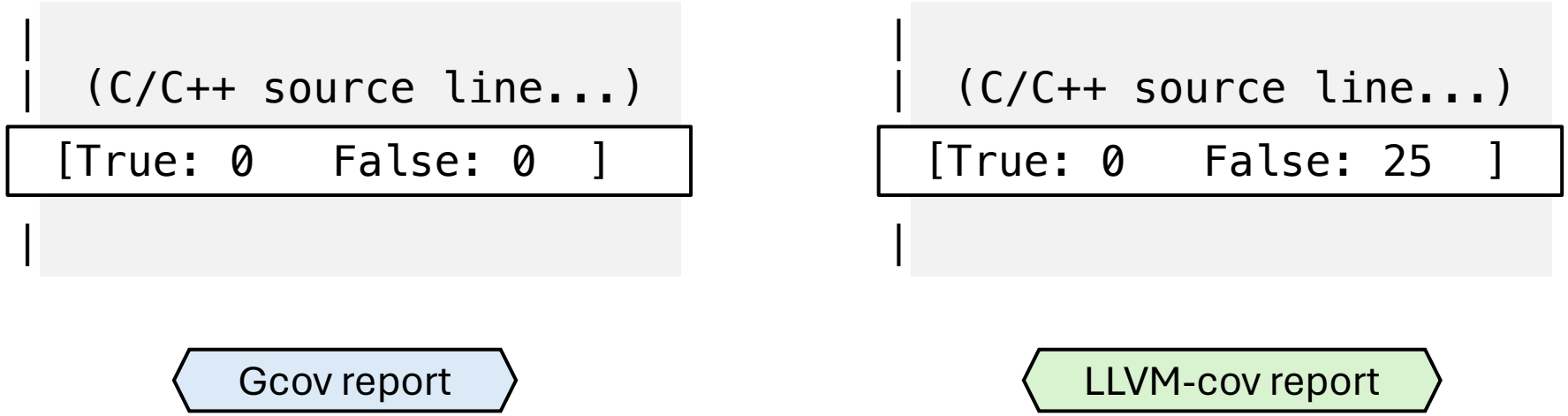

Number of branch conditions are the same: ✓
Move on to **value** comparison

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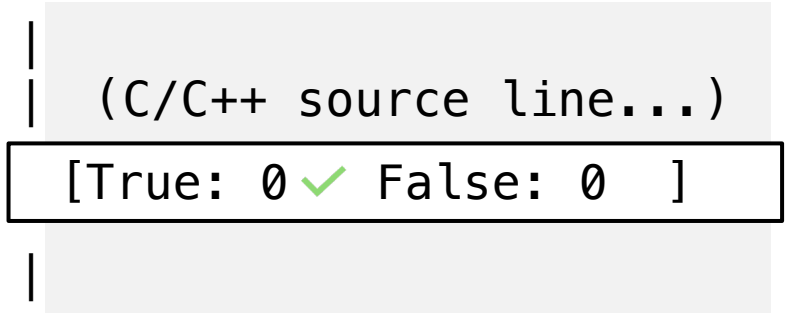

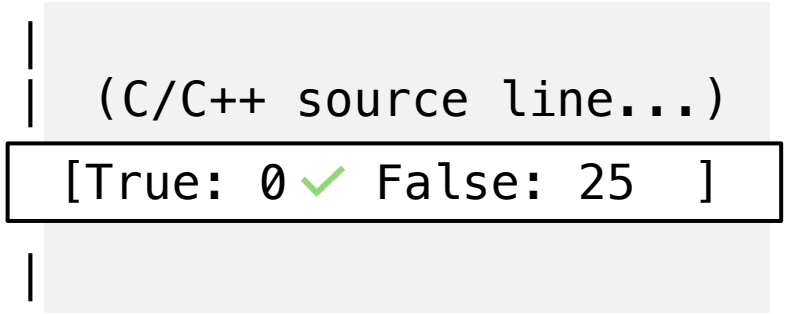

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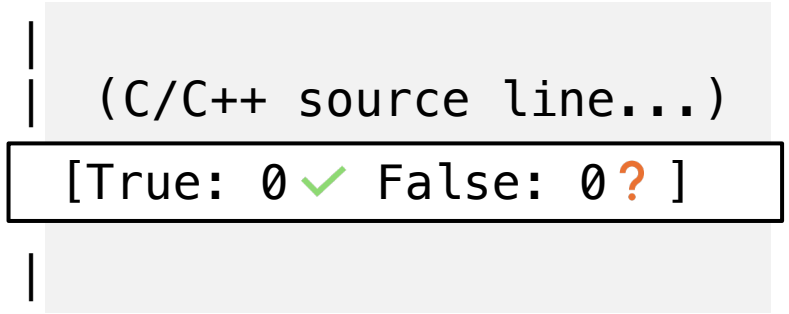

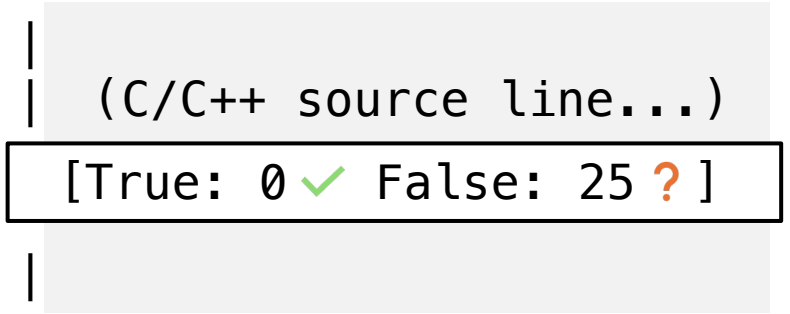

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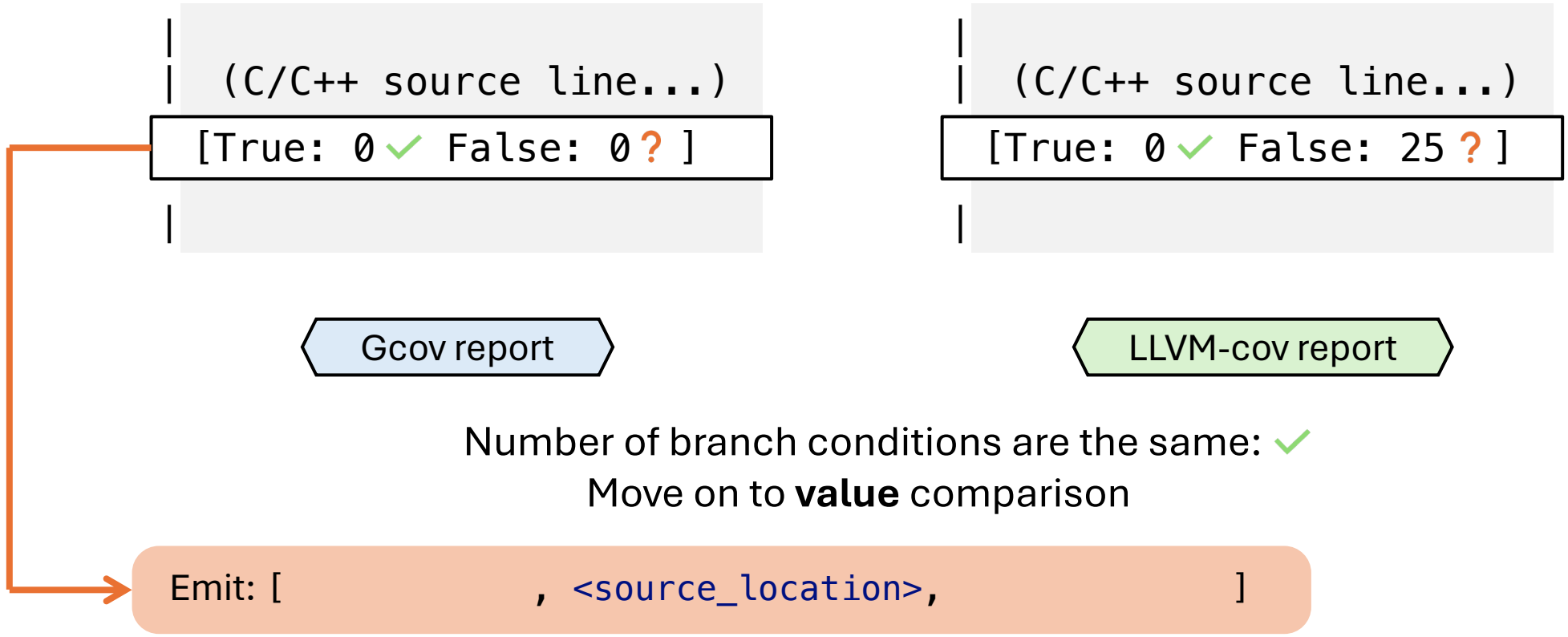

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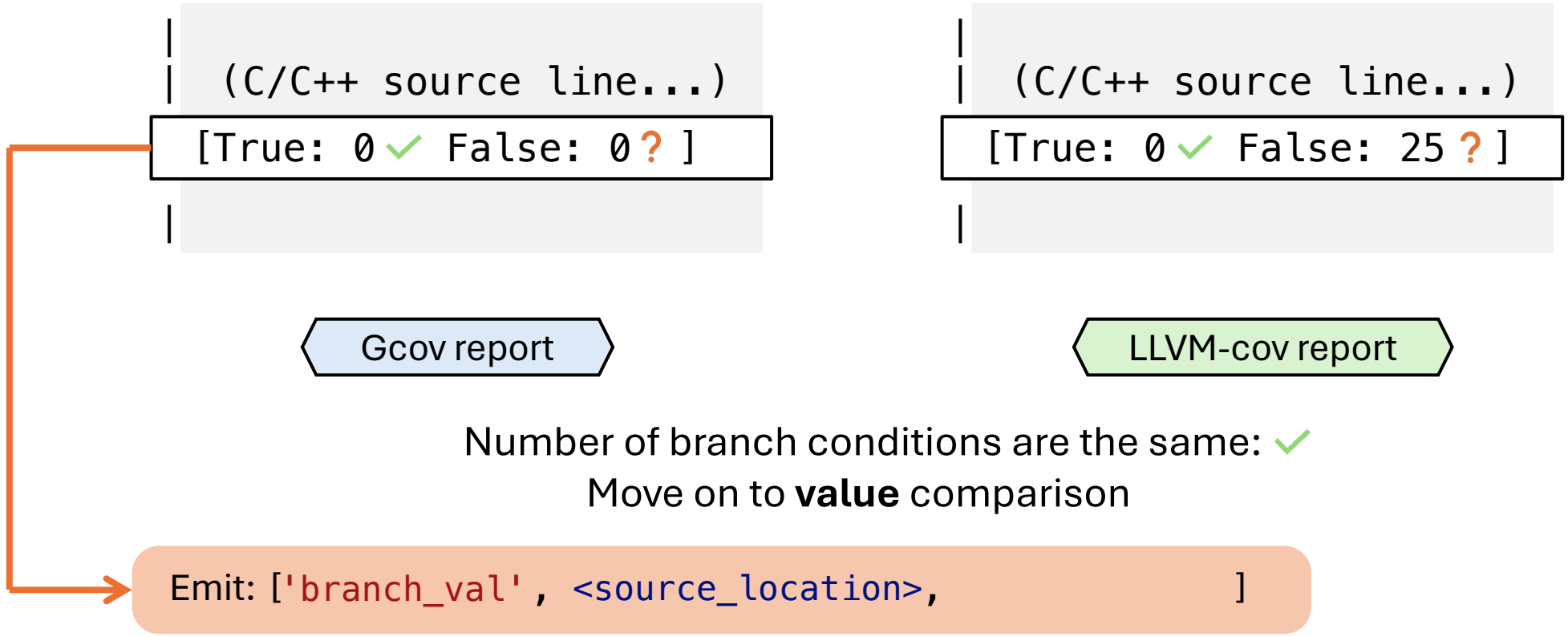

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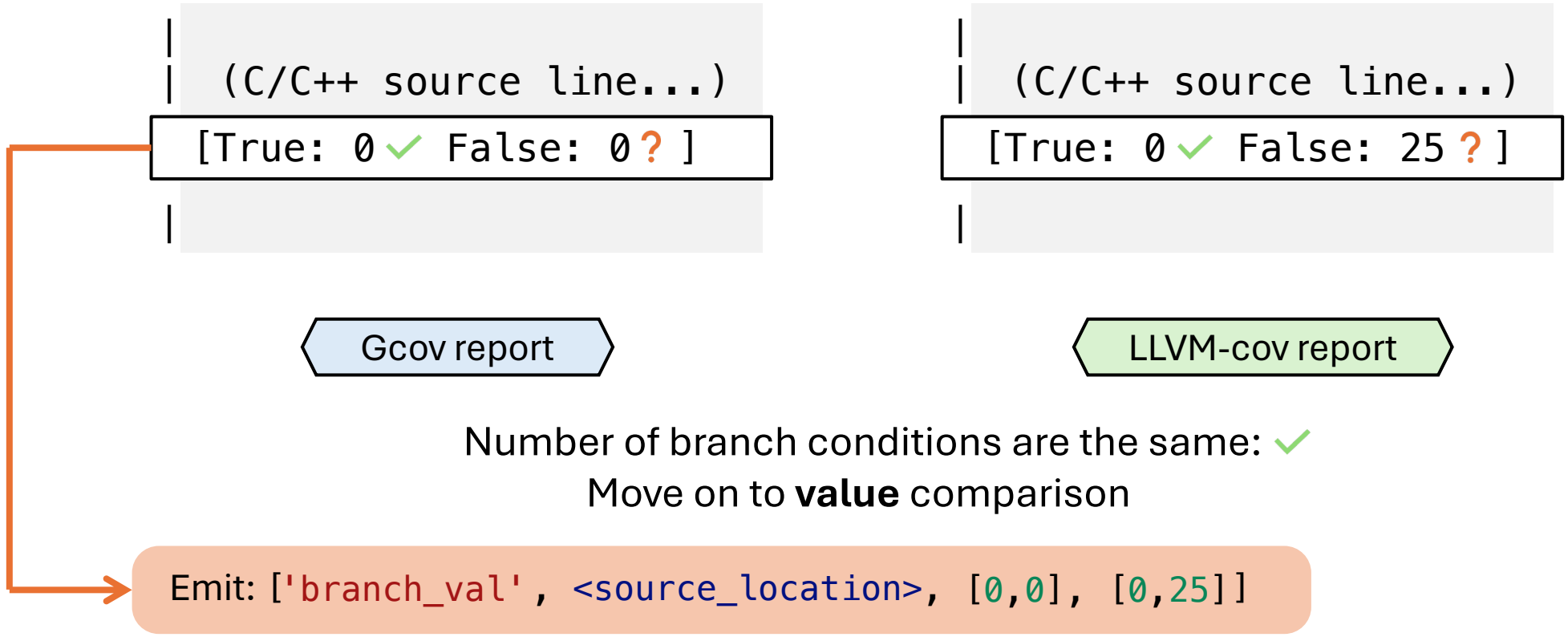

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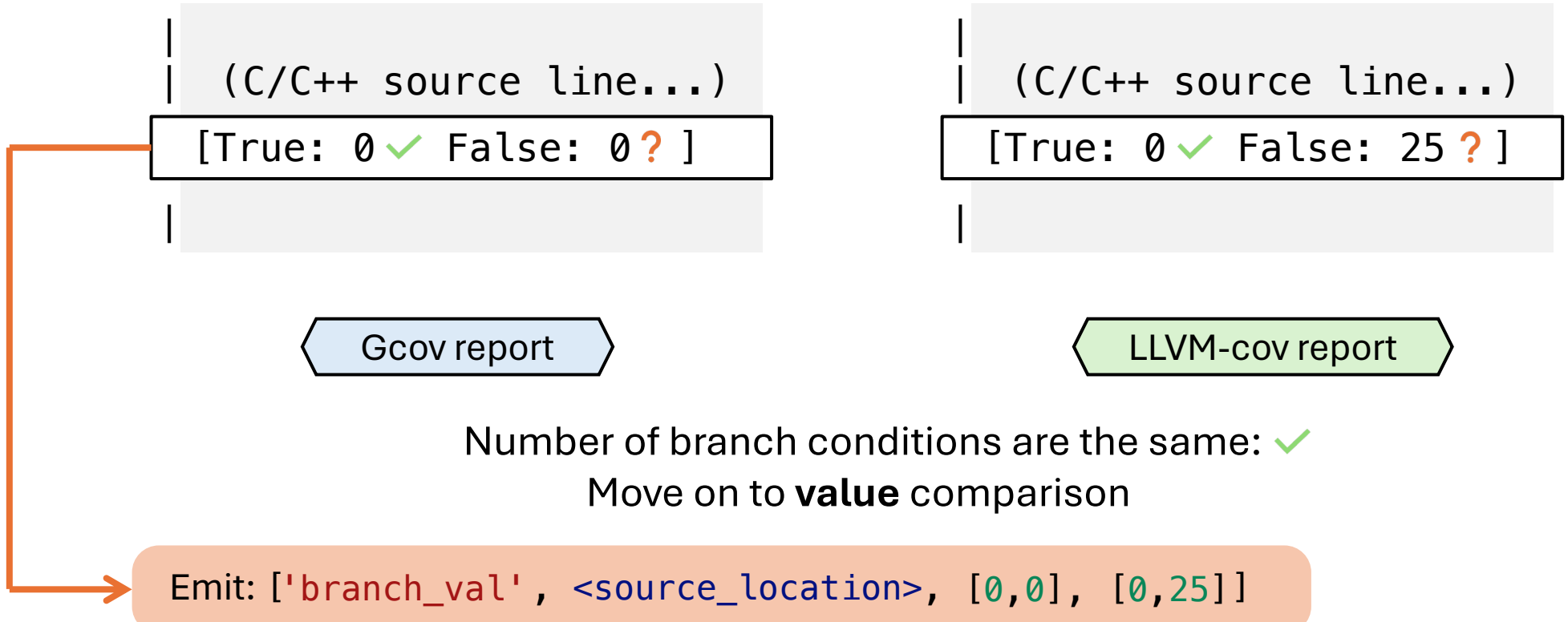

- MC/DC: `'mcdc_num'` and `'mcdc_val'`

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repeat = 1 `['line_val', 'src/ps/display.c:318', 6779, ..]`

repeat = 2 `['line_val', 'src/ps/display.c:318', 6878, ..]`

⋮

repeat = T `['line_val', 'src/ps/display.c:318', 6812, ..]`

Program Nondeterminism

- Even **the same tool can report different coverage** at different time or with different machine environment
  - Largely depending on package and tests

```
$ ps # part of "procps" package
    PID TTY          TIME CMD
2568258 pts/0    00:00:00 bash
2568281 pts/0    00:00:00 ps
```

- Countermeasure: hold off the alert until seeing stable inconsistency for repeated T runs ($T = 6$ for SC, 60 for ET)

repeat = 1 `['line_val', 'src/ps/display.c:318', 6779, ..]`

repeat = 2 `['line_val', 'src/ps/display.c:318', 6878, ..]`

⋮

repeat = T `['line_val', 'src/ps/display.c:318', 6812, ..]`

Program Nondeterminism

- Even **the same tool can report different coverage** at different time or with different machine environment
  - Largely depending on package and tests

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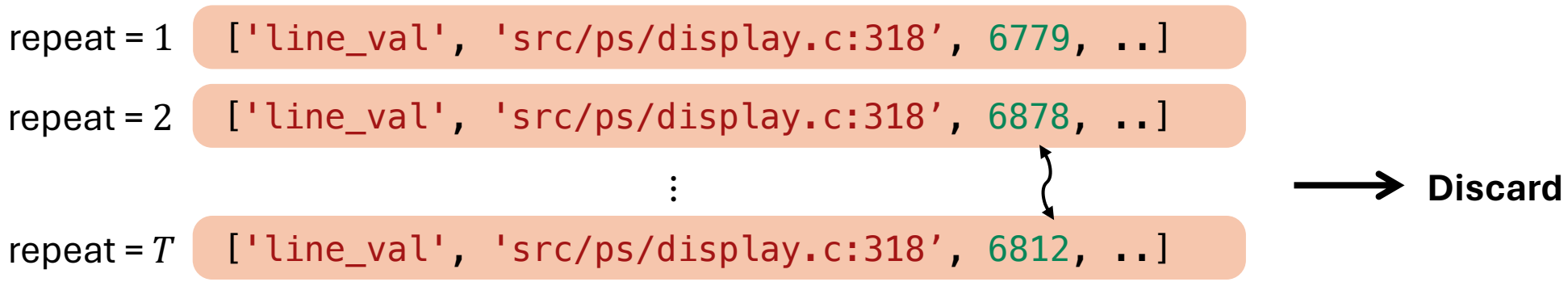
- Countermeasure: hold off the alert until seeing stable inconsistency for repeated T runs ($T = 6$ for SC, 60 for ET)



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$ ps # part of "procps" package
    PID TTY          TIME CMD
2568258 pts/0    00:00:00 bash
2568281 pts/0    00:00:00 ps
```

- Countermeasure: hold off the alert until seeing stable inconsistency for repeated T runs ($T = 6$ for SC, 60 for ET)

repeat = 1 `['line_val', 'src/ps/display.c:639', 462, 308]`

repeat = 2 `['line_val', 'src/ps/display.c:639', 462, 308]`

⋮

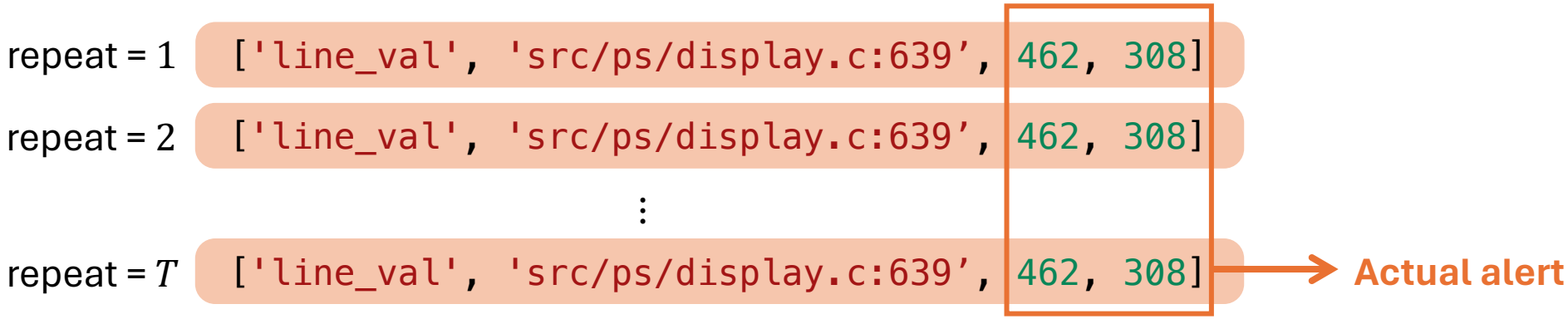
repeat = T `['line_val', 'src/ps/display.c:639', 462, 308]`

Program Nondeterminism

- Even **the same tool can report different coverage** at different time or with different machine environment
  - Largely depending on package and tests

```
$ ps # part of "procps" package
    PID TTY          TIME CMD
2568258 pts/0    00:00:00 bash
2568281 pts/0    00:00:00 ps
```

- Countermeasure: hold off the alert until seeing stable inconsistency for repeated T runs ($T = 6$ for SC, 60 for ET)



Program Nondeterminism

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2568258 pts/0    00:00:00 bash
2568281 pts/0    00:00:00 ps
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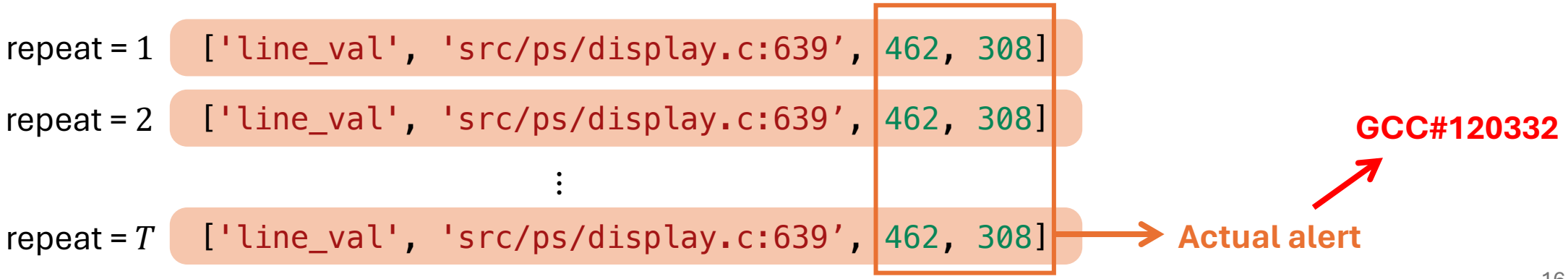
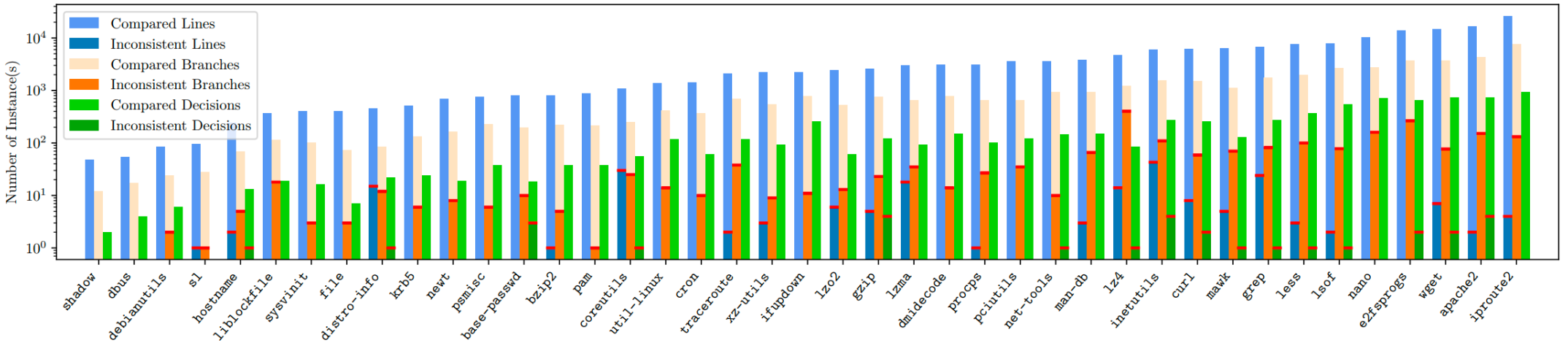
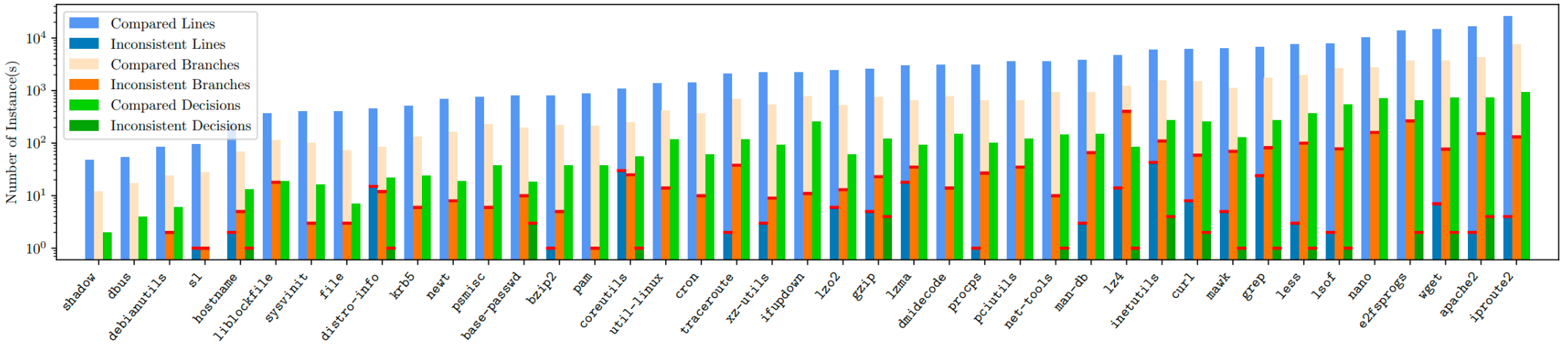


Inconsistencies in SC

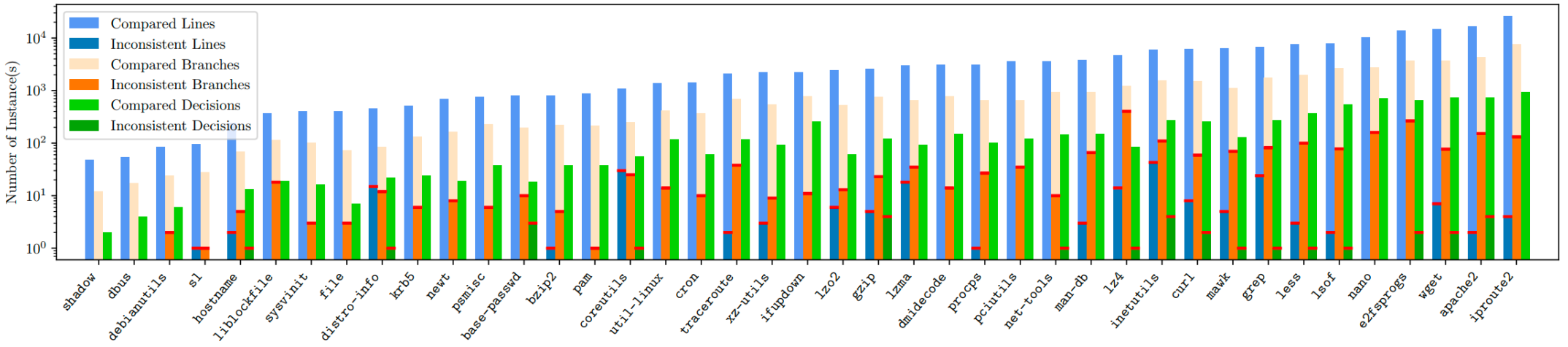


Inconsistencies in SC



	Lines	Branches	MC/DC
Total compared	168,549	44,126	7,527
Total inconsistent (*_num + *_val)	199	2,098 (1,884 + 214)	30 (29 + 1)

Inconsistencies in SC



	Lines	Branches	MC/DC
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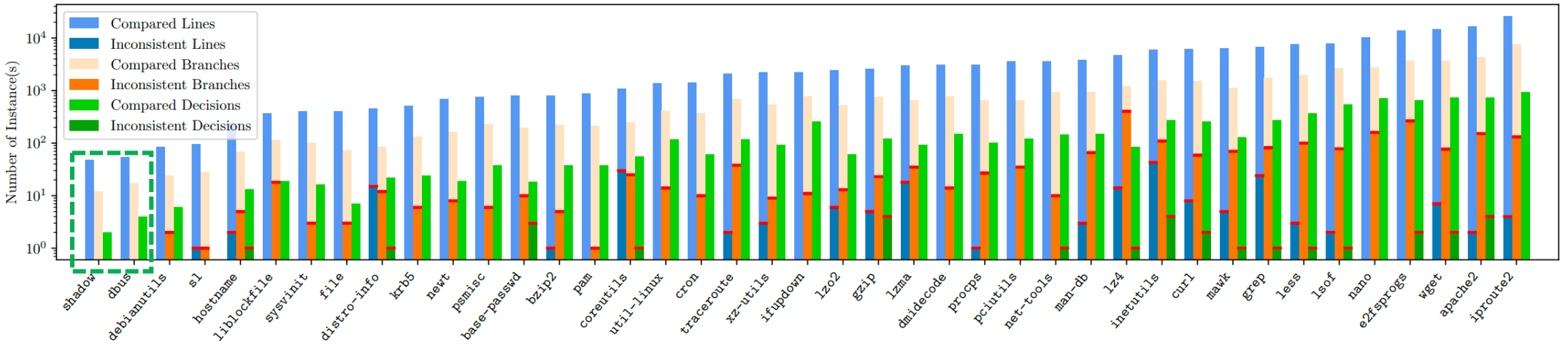
- **39** packages exhibit in total **2,327** inconsistencies

Inconsistencies in SC



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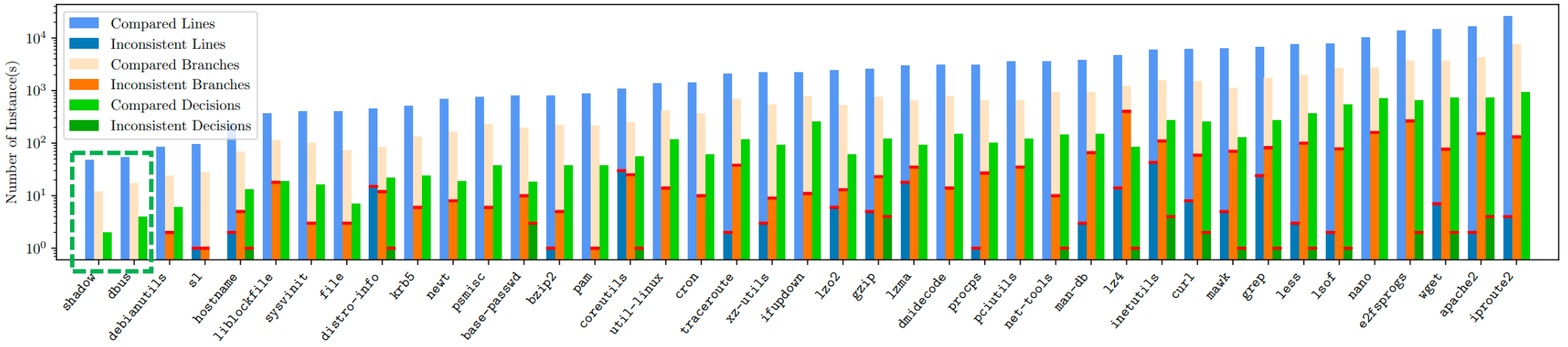

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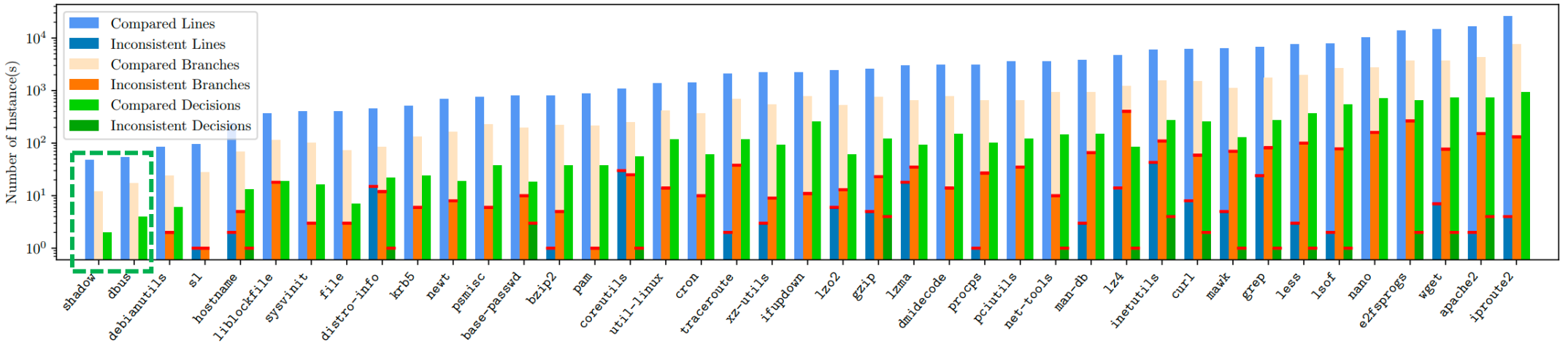
- **39** packages exhibit in total **2,327** inconsistencies
- Sampled **1,240** for inspection

Inconsistencies in SC



	Lines	Branches	MC/DC
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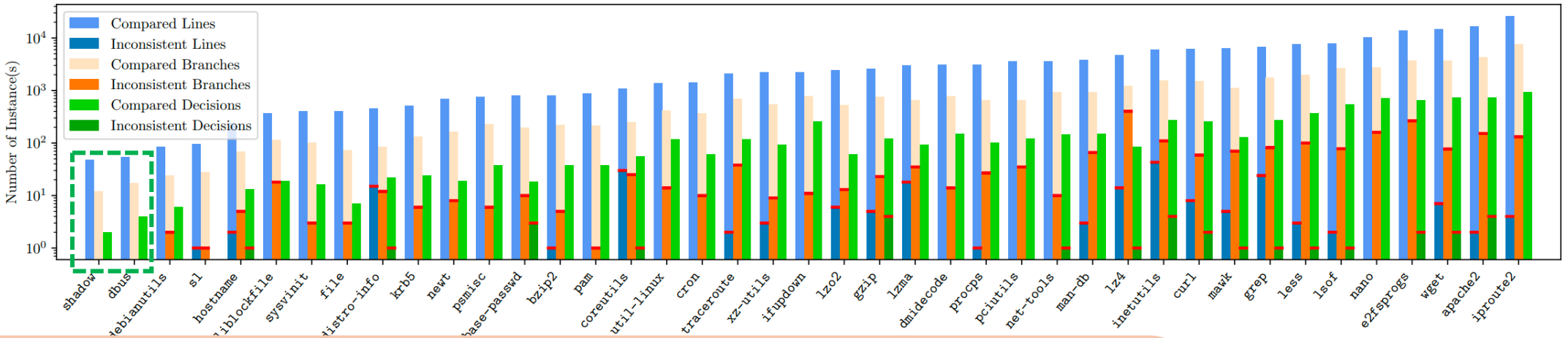
- **39** packages exhibit in total **2,327** inconsistencies
- Sampled **1,240** for inspection
  1. Bugs (256, detailed later)
  2. Conventions (954, detailed later)
  3. Compiler-specific differences (30)

Inconsistencies in SC




```
< -I/usr/include/p11-kit-1 -DHAVE_LIBGNUTLS -DNDEBUG -O0 --coverage
< -fcondition-coverage -DNO_SSLv2 -D_FILE_OFFSET_BITS=64 -g -Wall
---
> -I/usr/include/p11-kit-1 -DHAVE_LIBGNUTLS -DNDEBUG -O0
> -fprofile-instr-generate -fcoverage-mapping -fcoverage-mcdc
> -DNO_SSLv2 -D_FILE_OFFSET_BITS=64 -g -Wall
```

(*_num + *_val)	199	(1,884 + 214)	(29 + 1)

exhibit in total **2,327**
ies

40 for inspection

Bugs (256, detailed later)

Conventions (954, detailed later)

3. Compiler-specific differences (30)

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs

#	Bug ID	Tool	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	120321	Gcov	L	9 (`lzo2`)	6 + 11	constant, loop, no code, variable scope
2	120478	Gcov	L	2 (`lzo2`)	3 + 2	inline function, multistmt
3	117412	Gcov	L	3 (`lz4`)	3 + 1	dereference, function, multiline, type conversion
4	120482	Gcov	L	1 (`lz4`)	3 + n/a	`const` keyword, inline function
5	120490	Gcov	L	1 (`lz4`)	3 + n/a	inline function, `switch-case`
6	117415	Gcov	L	1 (`apache2`)	1 + n/a	dereference, function, `if`, loop, multiline
7	120348	Gcov	L	1 (`lzma`)	1 + n/a	C++, `extern "C"`, function, loop, `return`, variable scope
8	120484	Gcov	L	1 (`curl`)	1 + n/a	`if`, loop, variable scope
9	120489	Gcov	L	1 (`lz4`)	1 + n/a	`continue`, dereference, loop, variable scope
10	120491	Gcov	L	1 (`lzma`)	1 + n/a	C++, constructor, multiline
11	120492	Gcov	L	1 (`lzma`)	1 + n/a	C++ standard library, type conversion
12	120486	Gcov	L/BN	5 (`distro-info`)	6 + 19	multiline, ternary
13	120319	Gcov	L/BN	1 (`lzma`)	3 + n/a	C++, function, multiline, variable scope
14	120332	Gcov	L/BN/BV	5 (`lzo2`)	4 + 16	`if`, loop, no code, `switch-case`
15	120485	Gcov	BV	1 (`mawk`)	23 + 0	preprocessor directive, header
16	140427	LLVM-cov	L	3 (`mawk`)	2 + 2	`goto`, `if`, loop, macro, `return`
17	114622	LLVM-cov	L	1 (`sl`)	1 + n/a	`break`, header, macro
18	116884	LLVM-cov	L	1 (`grep`)	1 + n/a	`break`, constant, loop
19	105341	LLVM-cov	L/BV	1 (`hostname`)	1 + n/a	concurrency
20	116902	LLVM-cov	BN	1 (`lzma`)	3 + n/a	C++ standard library, function
21	101241	LLVM-cov	BN/MN	1 (`net-tools`)	2 + n/a	comma, constant, tautological compare, type conversion
22	131505	LLVM-cov	BN/BV/MN	1 (`hostname`)	1 + n/a	dereference, header, macro, type conversion
23	121914	Gcov	L	1 (`lzo2`)	0 + 4	inline function
24	121897	Gcov	L	2 (`ifupdown`)	0 + 3	function, variable scope
25	121901	Gcov	L	1 (`bzip2`)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (`mawk`)	0 + 2	function, `if`, pointer
27	121902	Gcov	L	1 (`ifupdown`)	0 + 1	`continue`
28	158003	LLVM-cov	L	1 (`mawk`)	0 + 4	macro, `switch-case`
29	158080	LLVM-cov	L	2 (`distro-info`, `mawk`)	0 + 4	`break`, `if`, `switch-case`
30	157959	LLVM-cov	L	2 (`lzo2`, `procps`)	0 + 2	`goto`, `if`
31	157946	LLVM-cov	L	1 (`distro-info`)	0 + 1	preprocessor directive, header, macro
32	157981	LLVM-cov	L	1 (`psmisc`)	0 + 1	loop, multiline

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs

#	Bug ID	Tool	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	120321	Gcov	L	9 (`lzo2`)	6 + 11	constant, loop, no code, variable scope
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3	117412	Gcov	L	3 (`lz4`)	3 + 1	dereference, function, multiline, type conversion
4	120482	Gcov	L	1 (`lz4`)	3 + n/a	`const` keyword, inline function
5	120490	Gcov	L	1 (`lz4`)	3 + n/a	inline function, `switch-case`
6	117415	Gcov	L	1 (`apache2`)	1 + n/a	dereference, function, `if`, loop, multiline
7	120348	Gcov	L	1 (`lzma`)	1 + n/a	C++, `extern "C"`, function, loop, `return`, variable scope
8	120484	Gcov	L	1 (`curl`)	1 + n/a	`if`, loop, variable scope
9	120489	Gcov	L	1 (`lz4`)	1 + n/a	`continue`, dereference, loop, variable scope
10	120491	Gcov	L	1 (`lzma`)	1 + n/a	C++, constructor, multiline
11	120492	Gcov	L	1 (`lzma`)	1 + n/a	C++ standard library, type conversion
12	120486	Gcov	L/BN	5 (`distro-info`)	6 + 19	multiline, ternary
13	120319	Gcov	L/BN	1 (`lzma`)	3 + n/a	C++, function, multiline, variable scope
14	120332	Gcov	L/BN/BV	5 (`lzo2`)	4 + 16	`if`, loop, no code, `switch-case`
15	120485	Gcov	BV	1 (`mawk`)	23 + 0	preprocessor directive, header
16	140427	LLVM-cov	L	3 (`mawk`)	2 + 2	`goto`, `if`, loop, macro, `return`
17	114622	LLVM-cov	L	1 (`sl`)	1 + n/a	`break`, header, macro
18	116884	LLVM-cov	L	1 (`grep`)	1 + n/a	`break`, constant, loop
19	105341	LLVM-cov	L/BV	1 (`hostname`)	1 + n/a	concurrency
20	116902	LLVM-cov	BN	1 (`lzma`)	3 + n/a	C++ standard library, function
21	101241	LLVM-cov	BN/MN	1 (`net-tools`)	2 + n/a	comma, constant, tautological compare, type conversion
22	131505	LLVM-cov	BN/BV/MN	1 (`hostname`)	1 + n/a	dereference, header, macro, type conversion
23	121914	Gcov	L	1 (`lzo2`)	0 + 4	inline function
24	121897	Gcov	L	2 (`ifupdown`)	0 + 3	function, variable scope
25	121901	Gcov	L	1 (`bzip2`)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (`mawk`)	0 + 2	function, `if`, pointer
27	121902	Gcov	L	1 (`ifupdown`)	0 + 1	`continue`
28	158003	LLVM-cov	L	1 (`mawk`)	0 + 4	macro, `switch-case`
29	158080	LLVM-cov	L	2 (`distro-info`, `mawk`)	0 + 4	`break`, `if`, `switch-case`
30	157959	LLVM-cov	L	2 (`lzo2`, `procps`)	0 + 2	`goto`, `if`
31	157946	LLVM-cov	L	1 (`distro-info`)	0 + 1	preprocessor directive, header, macro
32	157981	LLVM-cov	L	1 (`psmisc`)	0 + 1	loop, multiline

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs
  - In both tools: **20** in Gcov and **12** in LLVM-cov

#	Bug ID	Tool	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	120321	Gcov	L	2 (lzo2)	6 + 11	constant, loop, no code, variable scope
2	120478	Gcov	L	1 (lz4)	3 + 1	dereference, function, multiline, type conversion
3	117412	Gcov	L	3 (lz4)	3 + 1	dereference, function, multiline, type conversion
4	120482	Gcov	L	1 (lz4)	3 + n/a	`const` keyword, inline function
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6	117415	Gcov	L	1 (apache2)	1 + n/a	dereference, function, `if`, loop, multiline
7	120348	Gcov	L	1 (lzma)	1 + n/a	C++, `extern "C"`, function, loop, `return`, variable scope
8	120484	Gcov	L	1 (curl)	1 + n/a	`if`, loop, variable scope
9	120489	Gcov	L	1 (lz4)	1 + n/a	`continue`, dereference, loop, variable scope
10	120491	Gcov	L	1 (lzma)	1 + n/a	C++, constructor, multiline
11	120492	Gcov	L	1 (lzma)	1 + n/a	C++ standard library, type conversion
12	120486	Gcov	L/BN	5 (distro-info)	6 + 19	multiline, ternary
13	120319	Gcov	L/BN	1 (lzma)	3 + n/a	C++, function, multiline, variable scope
14	120332	Gcov	L/BN/BV	5 (lzo2)	4 + 16	`if`, loop, no code, `switch-case`
15	120485	Gcov	BV	1 (mawk)	23 + 0	preprocessor directive, header
16	140427	LLVM-cov	L	3 (mawk)	2 + 2	`goto`, `if`, loop, macro, `return`
17	114622	LLVM-cov	L	1 (sl)	1 + n/a	`break`, header, macro
18	116884	LLVM-cov	L	1 (grep)	1 + n/a	`break`, constant, loop
19	105341	LLVM-cov	L/BV	1 (hostname)	1 + n/a	concurrency
20	116902	LLVM-cov	BN	1 (lzma)	3 + n/a	C++ standard library, function
21	101241	LLVM-cov	BN/MN	1 (net-tools)	2 + n/a	comma, constant, tautological compare, type conversion
22	131505	LLVM-cov	BN/BV/MN	1 (hostname)	1 + n/a	dereference, header, macro, type conversion
23	121914	Gcov	L	1 (lzo2)	0 + 4	inline function
24	121897	Gcov	L	2 (ifupdown)	0 + 3	function, variable scope
25	121901	Gcov	L	1 (bzip2)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (mawk)	0 + 2	function, `if`, pointer
27	121902	Gcov	L	1 (ifupdown)	0 + 1	`continue`
28	158003	LLVM-cov	L	1 (mawk)	0 + 4	macro, `switch-case`
29	158080	LLVM-cov	L	2 (distro-info, mawk)	0 + 4	`break`, `if`, `switch-case`
30	157959	LLVM-cov	L	2 (lzo2, procps)	0 + 2	`goto`, `if`
31	157946	LLVM-cov	L	1 (distro-info)	0 + 1	preprocessor directive, header, macro
32	157981	LLVM-cov	L	1 (psmisc)	0 + 1	loop, multiline

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs

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1	120321	Gcov	L	2 (lzo2)	6 + 11	constant, loop, no code, variable scope
2	120478	Gcov	L	1 (lzma)	3 + 1	dereference, function, multiline, type conversion
3	117412	Gcov	L	3 (lz4)	3 + 1	dereference, function, multiline, type conversion
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12	120486	Gcov	L/BN	5 (distro-info)	6 + 19	multiline, ternary
13	120319	Gcov	L/BN	1 (lzma)	3 + n/a	C++, function, multiline, variable scope
14	120332	Gcov	L/BN/BV	5 (lzo2)	4 + 16	`if`, loop, no code, `switch-case`
15	120485	Gcov	BV	1 (mawk)	23 + 0	preprocessor directive, header
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17	114622	LLVM-cov	L	1 (sl)	1 + n/a	`break`, header, macro
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19	105341	LLVM-cov	L/BV	1 (hostname)	1 + n/a	concurrency
20	116902	LLVM-cov	BN	1 (lzma)	3 + n/a	C++ standard library, function
21	101241	LLVM-cov	BN/MN	1 (net-tools)	2 + n/a	comma, constant, tautological compare, type conversion
22	131505	LLVM-cov	BN/BV/MN	1 (hostname)	1 + n/a	dereference, header, macro, type conversion
23	121914	Gcov	L	1 (lzo2)	0 + 4	inline function
24	121897	Gcov	L	2 (ifupdown)	0 + 3	function, variable scope
25	121901	Gcov	L	1 (bzip2)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (mawk)	0 + 2	function, `if`, pointer
27	121902	Gcov	L	1 (ifupdown)	0 + 1	`continue`
28	158003	LLVM-cov	L	1 (mawk)	0 + 4	macro, `switch-case`
29	158080	LLVM-cov	L	2 (distro-info, mawk)	0 + 4	`break`, `if`, `switch-case`
30	157959	LLVM-cov	L	2 (lzo2, procps)	0 + 2	`goto`, `if`
31	157946	LLVM-cov	L	1 (distro-info)	0 + 1	preprocessor directive, header, macro
32	157981	LLVM-cov	L	1 (psmisc)	0 + 1	loop, multiline

- In both tools: **20** in Gcov and **12** in LLVM-cov
- Across all 3 metrics; one bug can affect multiple

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs

#	Bug ID	Tool	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	120321	Gcov	L	2 (lzo2)	6 + 11	constant, loop, no code, variable scope
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5	120490	Gcov	L	1 (lz4)	3 + n/a	inline function, `switch-case`
6	117415	Gcov	L	1 (lz4)	3 + n/a	inline function, `switch-case`
7	120348	Gcov	L	1 (lzma)	1 + n/a	C++, `extern "C"`, function, loop, `recurn`, variable scope
8	120484	Gcov	L	1 (curl)	1 + n/a	`if`, loop, variable scope
9	120489	Gcov	L	1 (lz4)	1 + n/a	`continue`, dereference, loop, variable scope
10	120491	Gcov	L	1 (lz4)	1 + n/a	C++, constructor, multiline
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12	120486	Gcov	L/BN	5 (distro-info)	6 + 19	multiline, ternary
13	120319	Gcov	L/BN	1 (lzma)	3 + n/a	C++, function, multiline, variable scope
14	120332	Gcov	L/BN/BV	5 (lzo2)	4 + 16	`if`, loop, no code, `switch-case`
15	120485	Gcov	BV	1 (mawk)	23 + 0	preprocessor directive, header
16	140427	LLVM-cov	L	3 (mawk)	2 + 2	`goto`, `if`, loop, macro, `return`
17	114622	LLVM-cov	L	1 (sl)	1 + n/a	`break`, header, macro
18	116884	LLVM-cov	L	1 (grep)	1 + n/a	`break`, constant, loop
19	105341	LLVM-cov	L/BV	1 (hostname)	1 + n/a	concurrency
20	116902	LLVM-cov	BN	1 (lzma)	3 + n/a	C++ standard library, function
21	101241	LLVM-cov	BN/MN	1 (net-tools)	2 + n/a	comma, constant, tautological compare, type conversion
22	131505	LLVM-cov	BN/BV/MN	1 (hostname)	1 + n/a	dereference, header, macro, type conversion
23	121914	Gcov	L	1 (lzo2)	0 + 4	inline function
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25	121901	Gcov	L	1 (bzip2)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (mawk)	0 + 2	function, `if`, pointer
27	121902	Gcov	L	1 (ifupdown)	0 + 1	`continue`
28	158003	LLVM-cov	L	1 (mawk)	0 + 4	macro, `switch-case`
29	158080	LLVM-cov	L	2 (distro-info, mawk)	0 + 4	`break`, `if`, `switch-case`
30	157959	LLVM-cov	L	2 (lzo2, procps)	0 + 2	`goto`, `if`
31	157946	LLVM-cov	L	1 (distro-info)	0 + 1	preprocessor directive, header, macro
32	157981	LLVM-cov	L	1 (psmisc)	0 + 1	loop, multiline

- In both tools: **20** in Gcov and **12** in LLVM-cov
- Across all 3 metrics; one bug can affect multiple
- Recurring patterns

Bugs Found by This Work

- **2** crashing bugs in LLVM-cov
- **32** wrong coverage bugs

#	Bug ID	Tool	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	120321	Gcov	L	2 (lzo2)	6 + 11	constant, loop, no code, variable scope
2	120478	Gcov	L	3 (lz4)	3 + 1	dereference, function, multiline, type conversion
3	117412	Gcov	L	3 (lz4)	3 + 1	dereference, function, multiline, type conversion
4	120482	Gcov	L	1 (lz4)	3 + n/a	`const` keyword, inline function
5	120490	Gcov	L	1 (lz4)	3 + n/a	inline function, `switch-case`
6	117415	Gcov	L	1 (lzma)	1 + n/a	C++, `extern "C"`, function, loop, `return`, variable scope
7	120348	Gcov	L	1 (lzma)	1 + n/a	C++, `extern "C"`, function, loop, `return`, variable scope
8	120484	Gcov	L	1 (curl)	1 + n/a	`if`, loop, variable scope
9	120489	Gcov	L	1 (lz4)	1 + n/a	`continue`, dereference, loop, variable scope
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25	121901	Gcov	L	1 (bsdmainutils)	0 + 3	dereference, loop, `malloc`, multiline
26	121896	Gcov	L	1 (ifupdown)	0 + 2	function, `if`, pointer
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Example Bugs

```
  |  unsigned char g, h;
1 |  int main(void) {
1 |     if (g < 256 && h)
    [True: 1, False: 0]
    [True: 0, False: 1]

        C1, C2   Result
    1 { T , F  = F     }

    C1-Pair: not covered
    C2-Pair: not covered

0 |         return 1;
1 |  }
```

Input program for LLVM#101241,
reduced from `net-tools` package

Example Bugs

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  |  unsigned char g, h;
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```

```
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Input program for LLVM#101241,
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```
test.c:3:11: warning: result of comparison of constant
 256 with expression of type 'unsigned char' is always
 true [-Wtautological-constant-out-of-range-compare]
    3 |      if (g < 256 && h)
      |          ~ ^ ~~~
```

Example Bugs

```
  |  unsigned char g, h;
1 |  int main(void) {
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```

~~[True: 1, False: 0]~~
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```
1 |  int main(void) {
  |      int i = 0;
1 |      while (1) {
6 |          0;
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1 |              break;
  |
  |      }
  |  }
```

Input program for GCC#120321, reduced from `lzo2` package

Example Bugs

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  |  unsigned char g, h;
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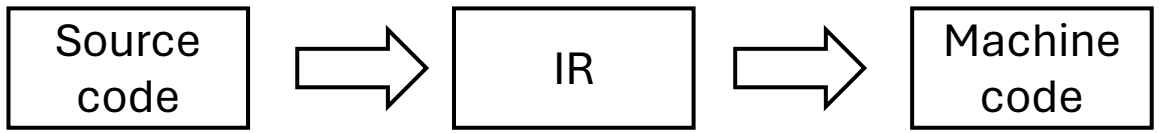

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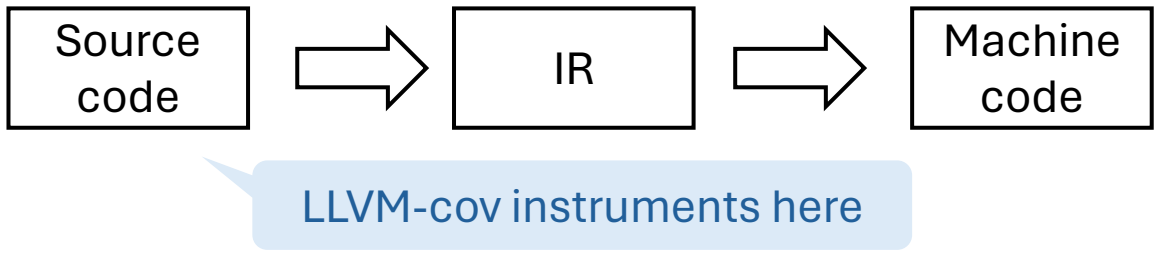

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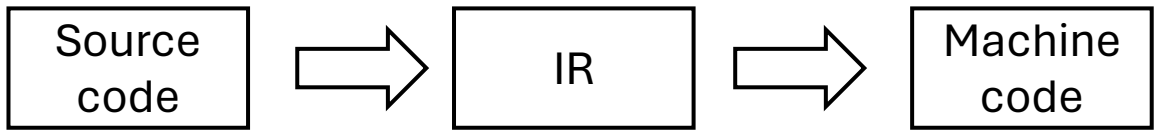

LLVM-cov instruments here

No sophisticated analysis

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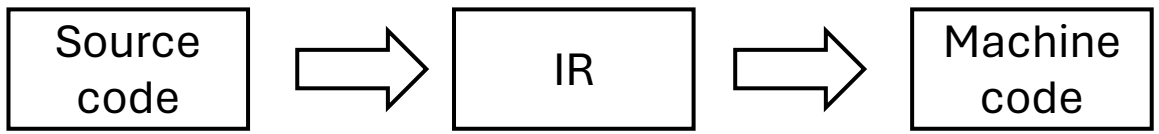

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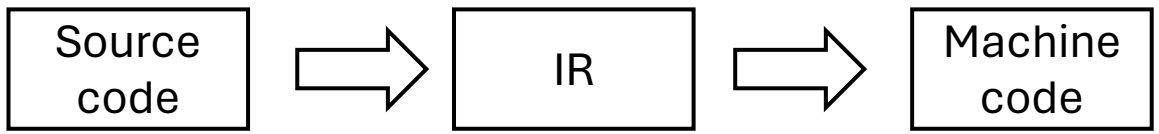

Gcov instruments here

Confused by source code that doesn't end up in the binary

Convention Differences

- Benign differences where both tools make their sense

#	Short Description	Incons. Type	Affected Package(s)	Occur.	Triggering Condition(s)
1	Complex control flow in macros	L	4 (`inetutils`, `lzma`)	9	`goto`, `if`, loop, macro
2	`do-while` statements	L	5 (`hostname`)	6	`if`, loop
3	`else if`	L	2 (`gzip`)	3	`if`
4	Multiline statements	L/BN/BV	19 (`iproute2`)	283	multiline, ternary
5	`switch-case` statements	L/BN/BV	12 (`mawk`)	142	no code, `switch-case`
6	Macros in decisions	BN	11 (`apache2`)	150	macro
7	Decisions in macros	BN	12 (`lz4`)	141	macro
8	Branches in inline functions	BN	1 (`lz4`)	118	inline function
9	Branches for assertions	BN	6 (`lz4`)	52	assertion
10	System header files	BN	7 (`iproute2`)	44	header
11	Constant condition	BN/MN	11 (`iproute2`)	58	constant
12	MC/DC ternary	MN	8 (`inetutils`)	12	ternary
13	MC/DC empty “then” clause	MN	3 (`gzip`)	6	`if`, no code
14	MC/DC inline function	MN	1 (`lz4`)	1	`const` keyword, inline function
15	Masking vs. unique-cause MC/DC	MV	1 (`distro-info`)	1	mixed conjunctive and disjunctive

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  - **Example 1:** line coverage for `switch`-`case` statements

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- Benign differences where both tools make their sense

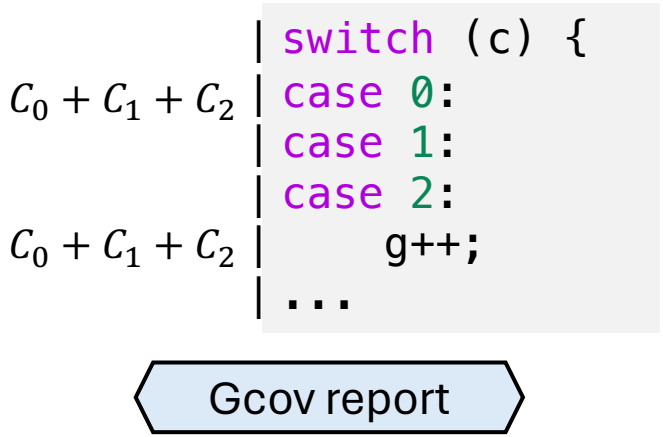
- **Example 1:** line coverage for `switch-case` statements

```
switch (c) {
case 0:
case 1:
case 2:
    g++;
...
```

Convention Differences

- Benign differences where both tools make their sense

- **Example 1:** line coverage for `switch-case` statements



Convention Differences

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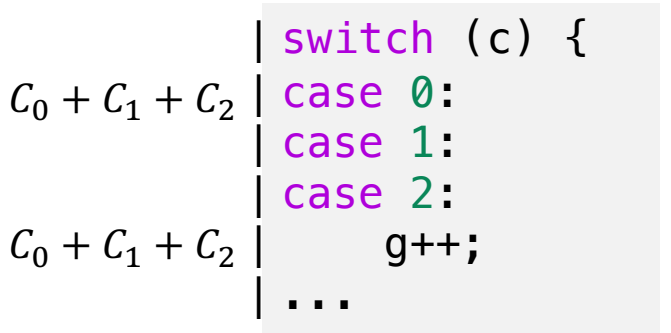
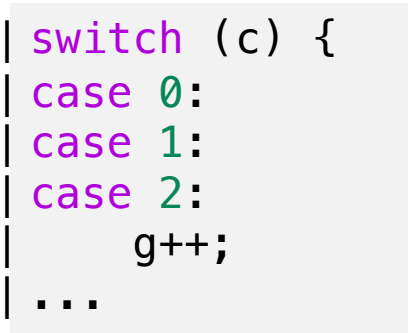


Gcov report



LLVM-cov report

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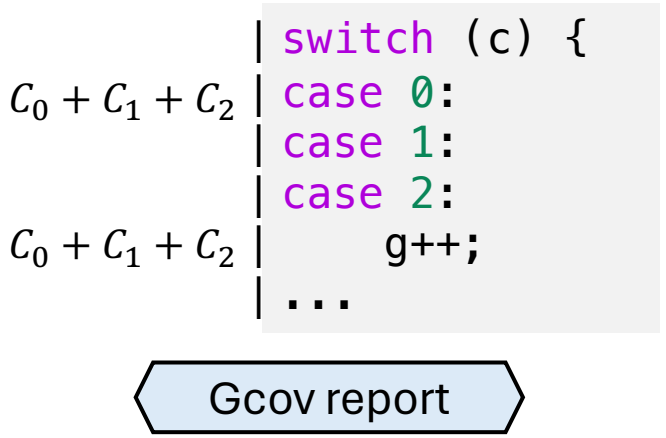

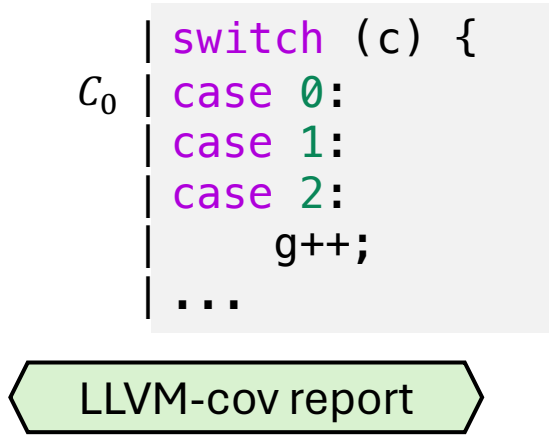

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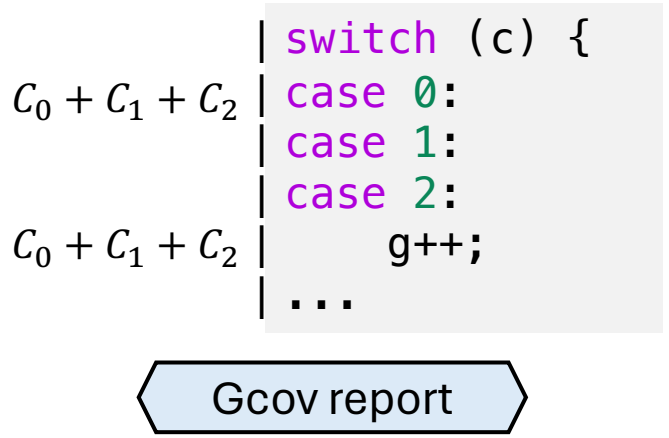

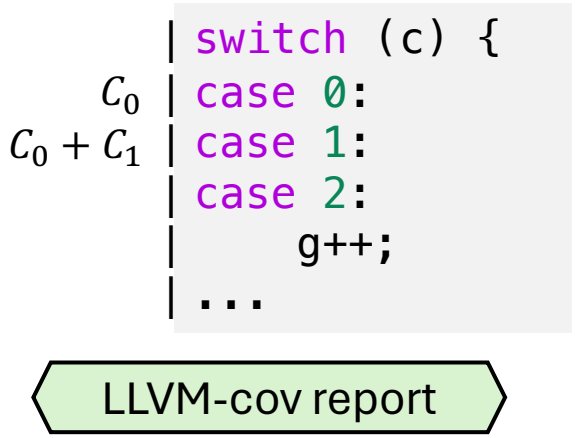

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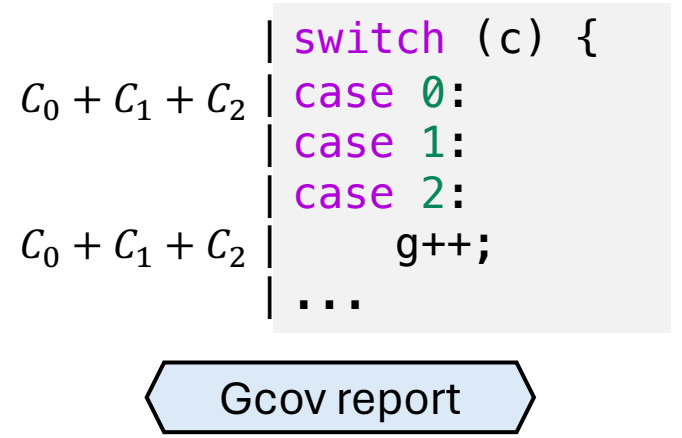

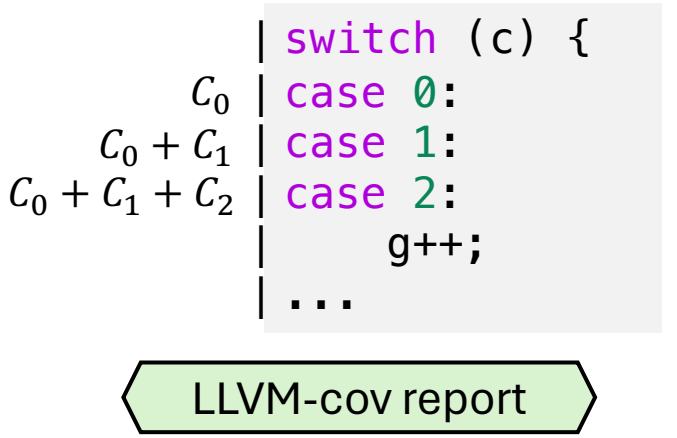

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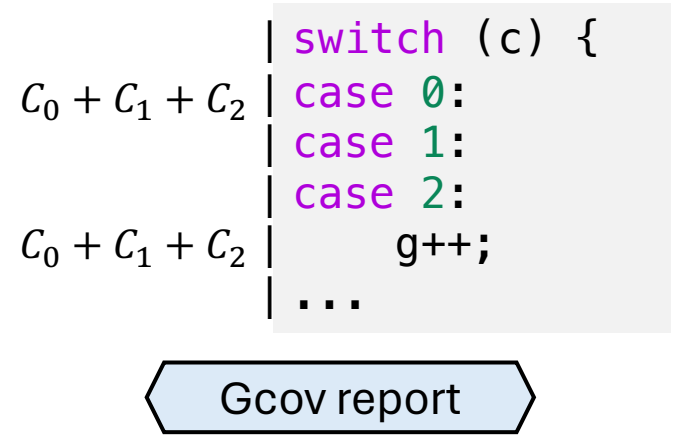

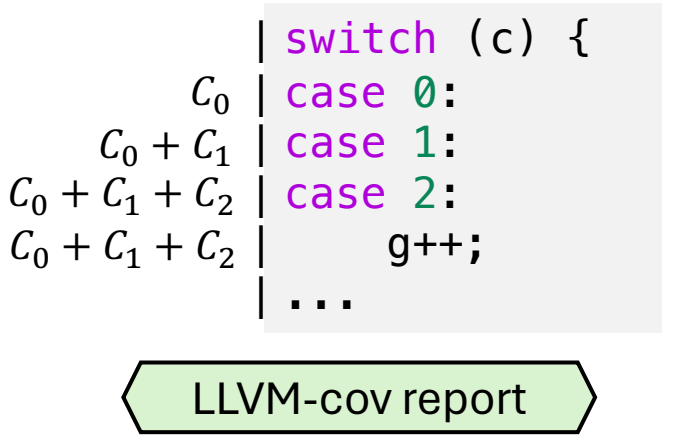

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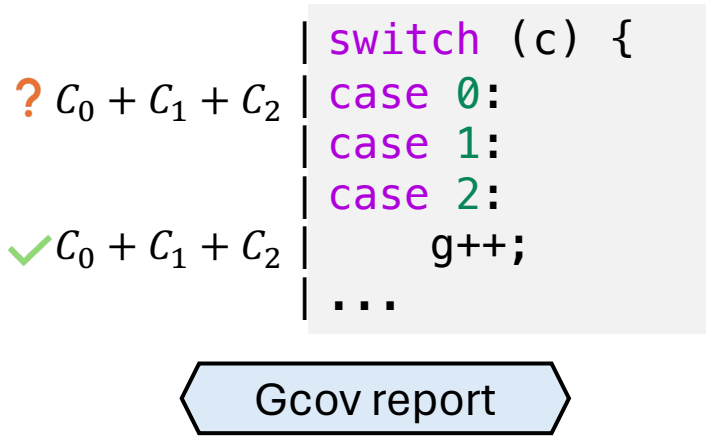

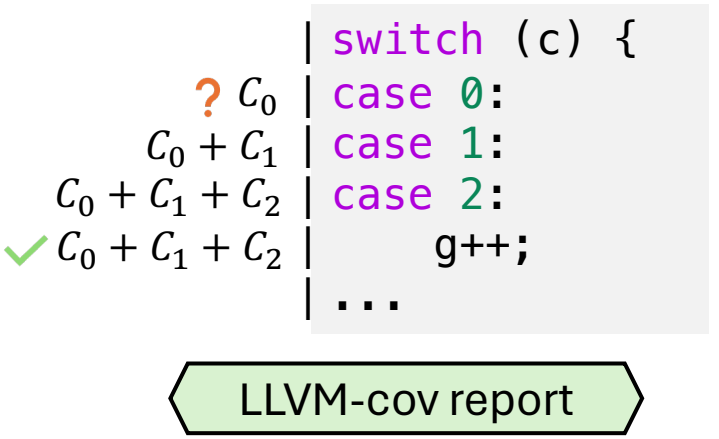

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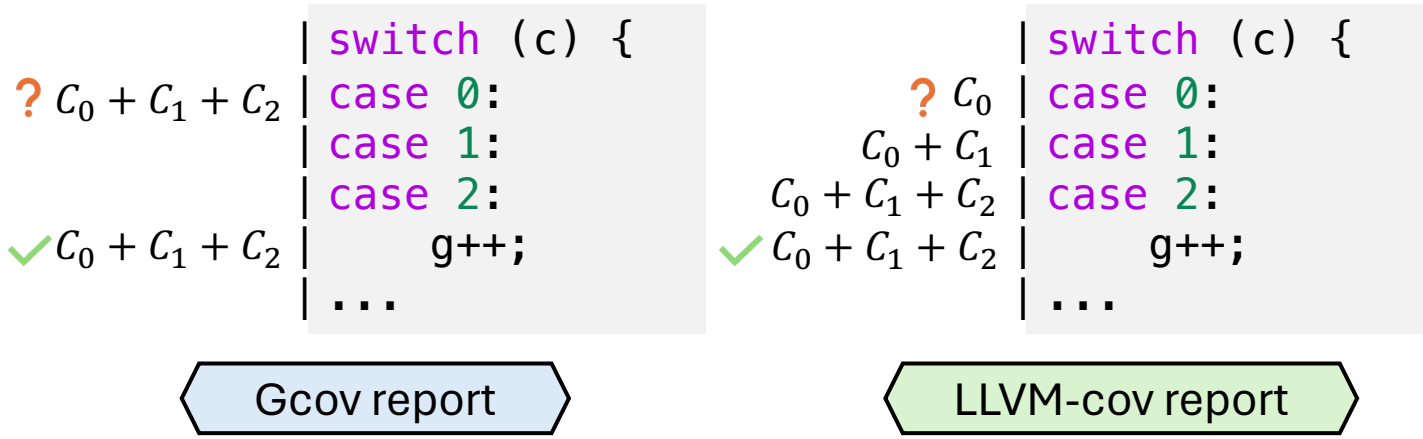

- **Example 2:** LLVM-cov can under-report MC/DC for certain expressions

Convention Differences

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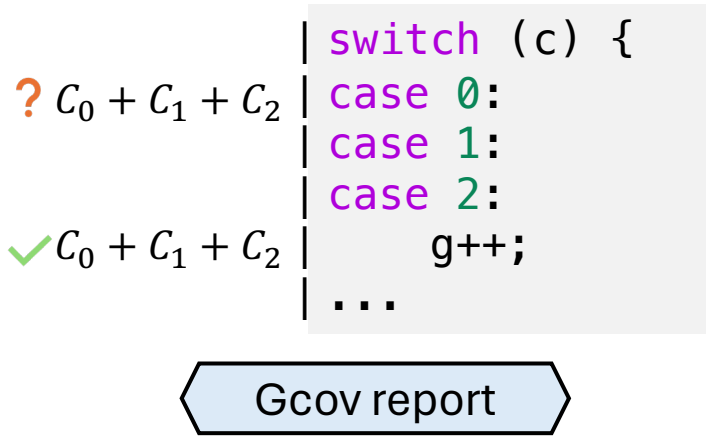

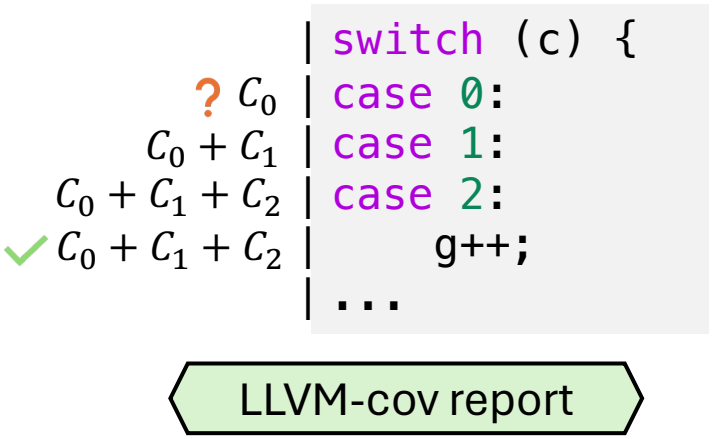

- **Example 2:** LLVM-cov can under-report MC/DC for certain expressions

It turns out Gcov and LLVM-cov adopt **two variants** of MC/DC definition, i.e., masking MC/DC and unique-cause MC/DC [Chilenski, 2001]

Convention Differences

- Benign differences where both tools make their sense

- **Example 1:** line coverage for `switch`-`case` statements

```
switch (c) {
case 0:
```

? $C_0 + C_1 + C_2$

? C_0

✓ $C_0 + C$

Developer clarification
https://discourse.llvm.org/t/rfc-source-based-mc-dc-code-coverage/59244/15

evodius96 1 whentojump Jun 2024

Hi @whentojump !

> - Q: Can you please confirm LLVM is currently following the unique-cause criteria?

I believe this is correct, as far as I understand; what I implemented in clang follows what other vendors have supported, especially LDRA. There aren't any improvements to mask out conditions using boolean logic evaluation of whole subexpressions. However, in the design doc, I inaccurately described it as *masking*, but this only applies to the fact that due to the short-circuit behavior of logical operators, unevaluatable conditions are masked-out and counted as dont-cares.

- **Example 2:** LLVM-cov can under-report MC/DC for certain expressions

Convention Differences

- Benign differences where both tools make their sense

- **Example 1:** line coverage for `switch`-`case` statements

? $C_0 + C_1 + C_2$

```
switch (c) {
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✓ $C_0 + C$

? C_0

```
switch (c) {
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```

Developer clarification
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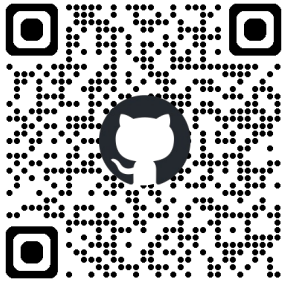
cares.

A standard on code coverage (just like C/C++ standard) is much desired

Summary

- Real-world projects help reveal unique coverage tool bugs
- DEBCOVDIFF: a differential testing framework
- Revealed 34 new bugs in Gcov and LLVM-cov
- Open sourced at https://github.com/xlab-uiuc/DEBCOVDIFF



Future Directions

- More challenging input than Debian packages, such as Linux kernel, Xen hypervisor
- Understand input dynamics using the calibrated coverage